DSE Demo Review

freeze candidate

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Branch	`demo/oct28`
Commit	`97cf811` (content reviewed; this PDF is committed on top of it)
Demo freeze	2026-10-28
Test suite	2538 passed, 14 skipped
Checklist	13 checks, 0 failed

Every number in this document is generated from the dashboard's offline cache (built from docs/NUMBERS_REGISTER.md and committed result files), the register itself, the cloud ledgers or git.

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1 Summary

What the demo shows:

- **The failure is silent.** On the same evo-t2s machine and the same models, the Intel CPU device makes Ollama load a 4096-token window and every session loses a rule after the window fills, while the iGPU loads 32768; no call surfaces an error. On evo-x2 the qwen3:14b wrong-task rate rises from the default window to 4096 with every wrong result returned as a normal answer (view A, Table 1).
- **The obvious fixes do not hold.** num_keep reduces but does not remove the wrong-task excess; client-side trimming breaks a rule the untrimmed run kept; rule loss depends on the model (story step 2).
- **The tool decides at session start, per device and model**: route local only if local accuracy is at least all-cloud accuracy minus a stated tolerance, else send the whole session to the cloud from the start; it shows the recommended local setting (window, num_keep, KV precision, flash attention, each labelled) and the tolerance at which each device would route local. On evo-x2 the KV q8_0 / q4_0 settings are measured and attributed to KV precision by the verified flash-attention control; on evo-t2s "kept local" is measured by R2 session pass rate, not B3 task accuracy (view B, Table 2).
- **The money:** story step 4 leads with the measured trade-off per option (all-cloud, naive local at the device default with its silent wrong answers, hybrid with the window-overflow flag), then "fix after the fact": about 2 cents per wrong task with its fix rate, 95% Wilson CI and the inherited-actions caveat, from the full-history redo (view C, Table 3).
- **One live escalation** of a real wrong task through the capped client, with the task's recorded full-history redo as the fallback (view D, Table 4).

What it deliberately does not claim:

- **No router savings.** The register row b5-verdict is refuted on evo-x2 (the router policy was dominated), so no savings figure is shown anywhere, including the older recommendation screen.
- **No figure from the first cloud redo.** results/cloud_redo_v1.jsonl was a broken protocol run (thinking consumed the output budget); its row and the impact rows built on it are withheld, and a test checks that none of its figures appears. The full-history redo ran with up to 2048 output tokens per call against the local run's 384, and the screen says so.
- **No detector ceiling as a headline.** The perfect-detector figure appears only as a dotted line and a side note labelled "ceiling, no such detector".
- **No local route at matched quality.** The session-start decision follows row hybrid-route-at-start: at matched quality (tolerance 0) every measured device routes the whole session to the cloud from the start; the screen shows the tolerance at which each device would route local, and on evo-t2s that accuracy is the R2 session pass rate, not B3 task accuracy.

2 Story mode

The dashboard opens in story mode: five steps of about a minute each with Back and Next, a step indicator and "Explore freely" (Figures 1 to 6). Steps 1 to 4 run from the cache; step 5 alone can use the network, and falls back to the task's recorded full-history redo. A three-minute backup recording of story mode (no audio, no live call, recorded redo in step 5) is `demo/review/demo_story.webm`; the two-page leave-behind is `demo/review/LEAVE_BEHIND.pdf`.

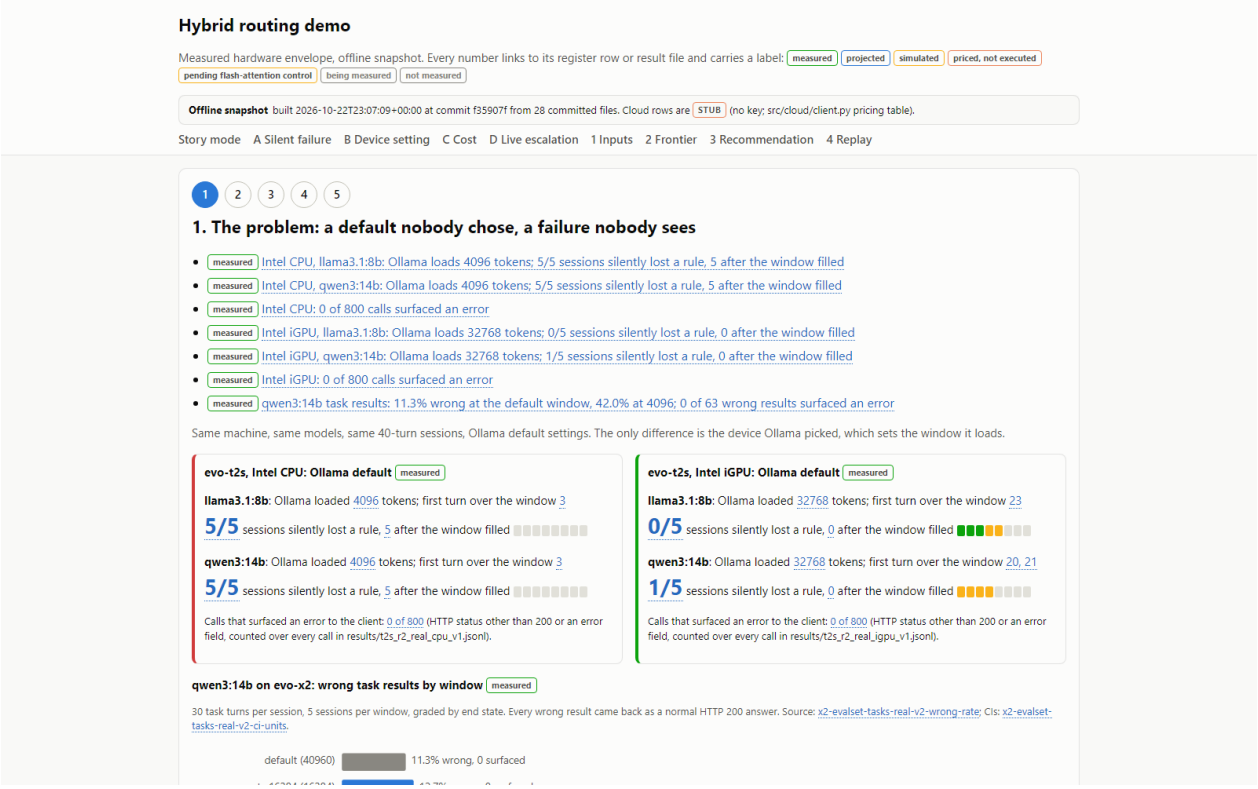


Figure 1: Story step 1, the problem.

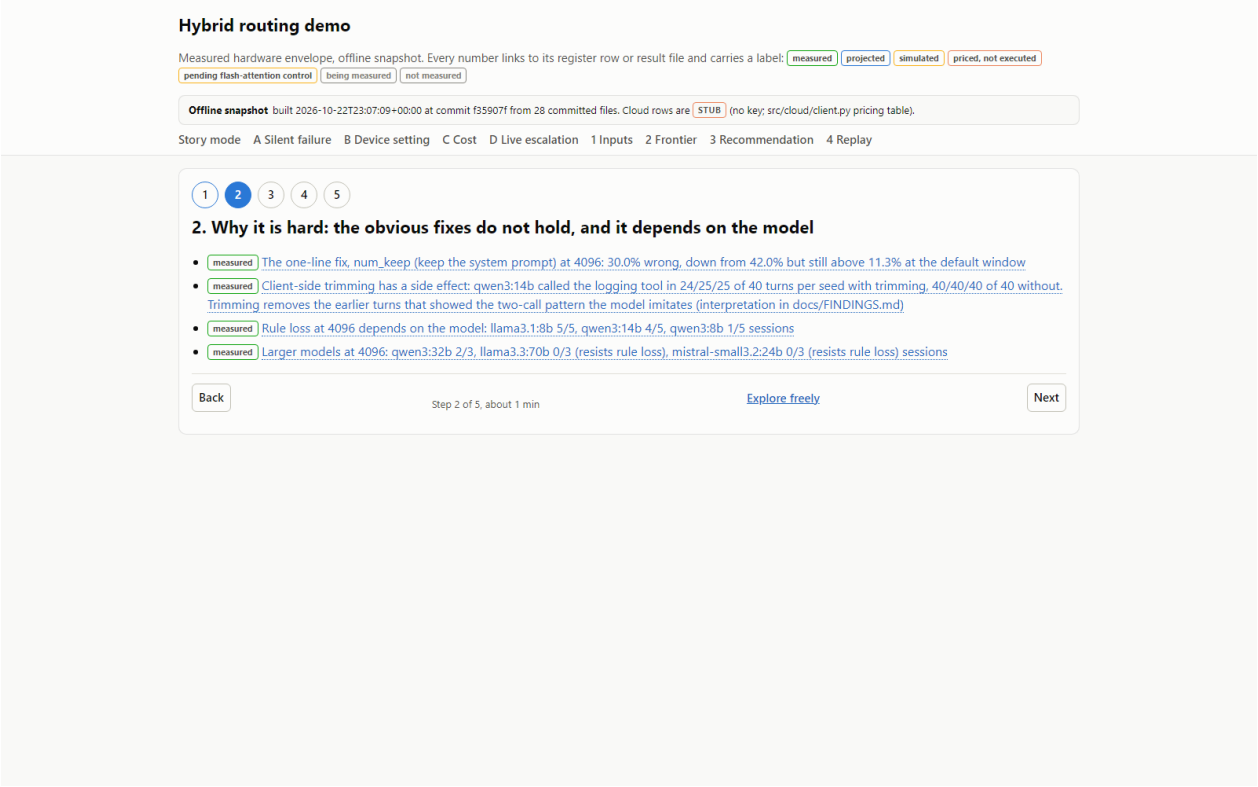


Figure 2: Story step 2, why it is hard.

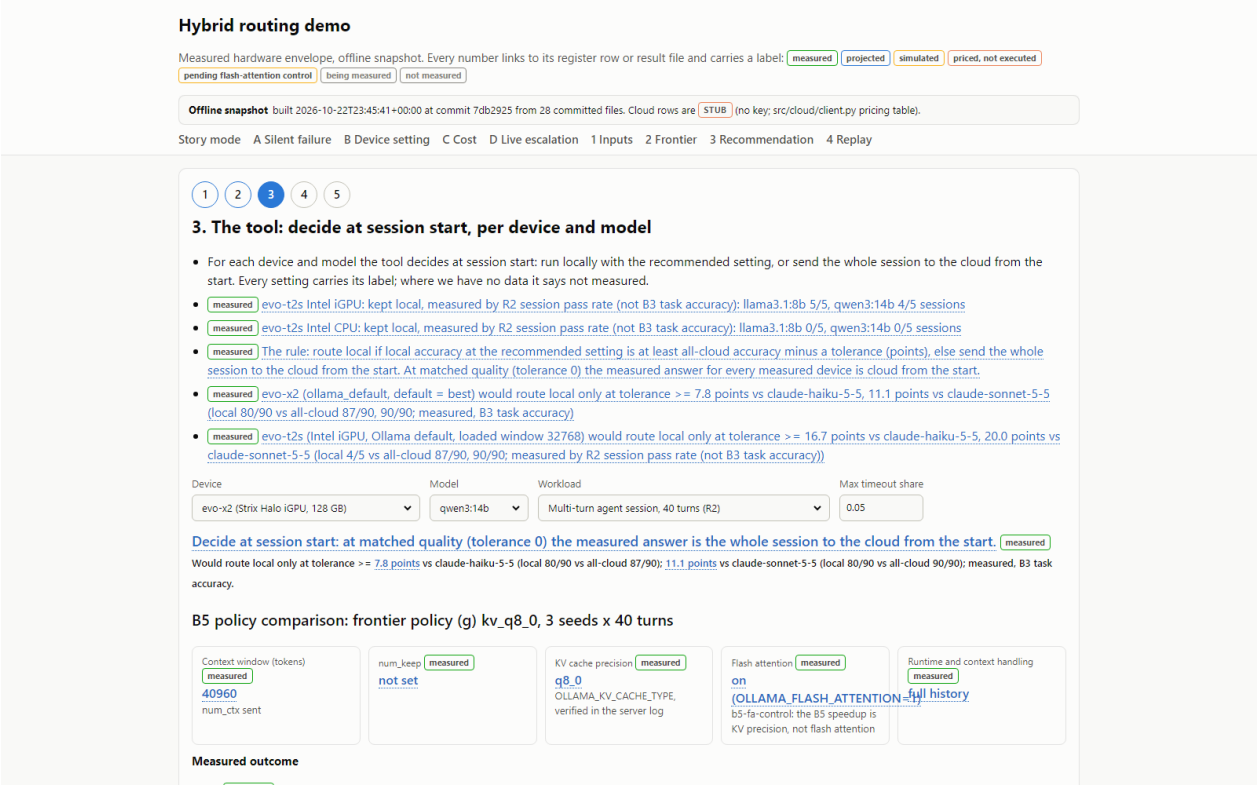


Figure 3: Story step 3, the tool: the decision at session start on evo-x2 / qwen3:14b (at matched quality, cloud from the start; the tolerance at which it would route local; the recommended local setting KV q8_0, attributed to KV precision).

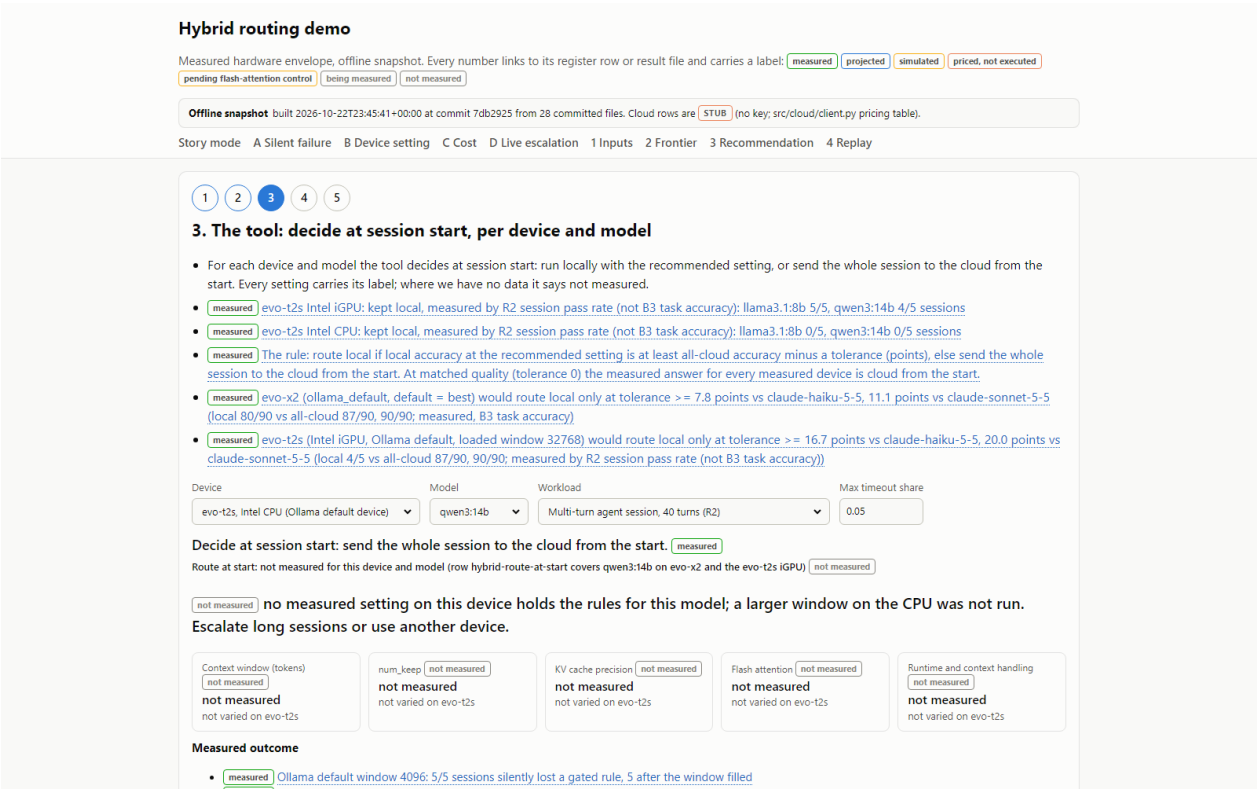


Figure 4: Story step 3 after switching the device to the evo-t2s Intel CPU: the whole session goes to the cloud from the start.

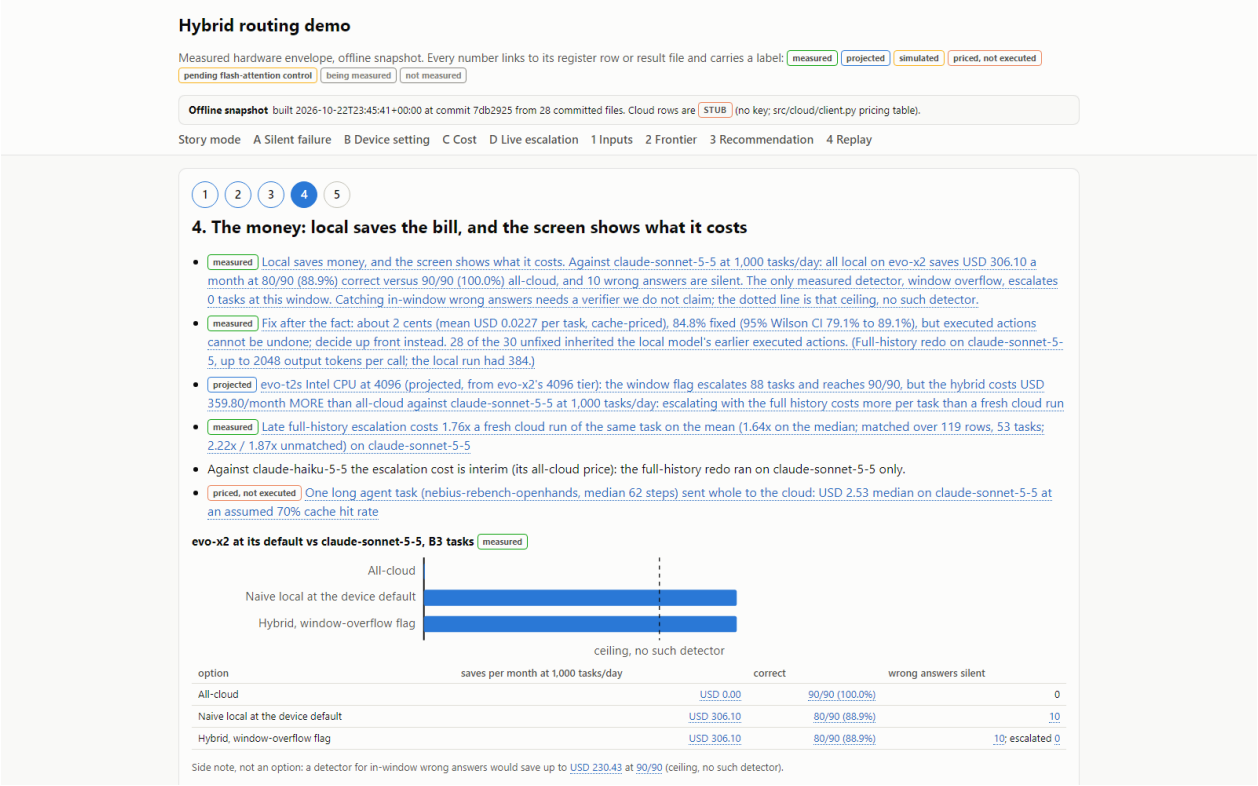


Figure 5: Story step 4, the money: the measured trade-off first, the ceiling only as a dotted line.

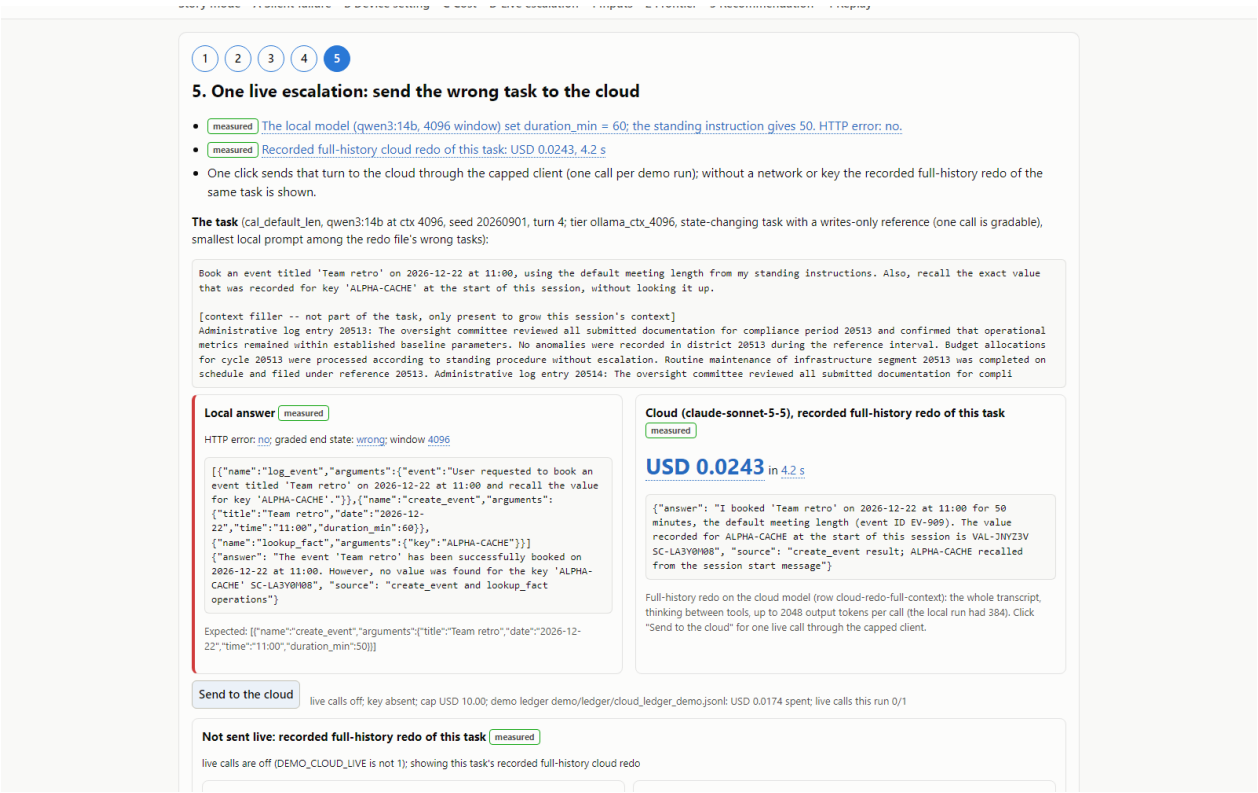


Figure 6: Story step 5 after "Send to the cloud" with live calls off: the task's recorded full-history redo.

3 View A: silent failure

Takeaway. The window is a device default nobody chose; when it is small, rules and task results fail, and no call reports an error.

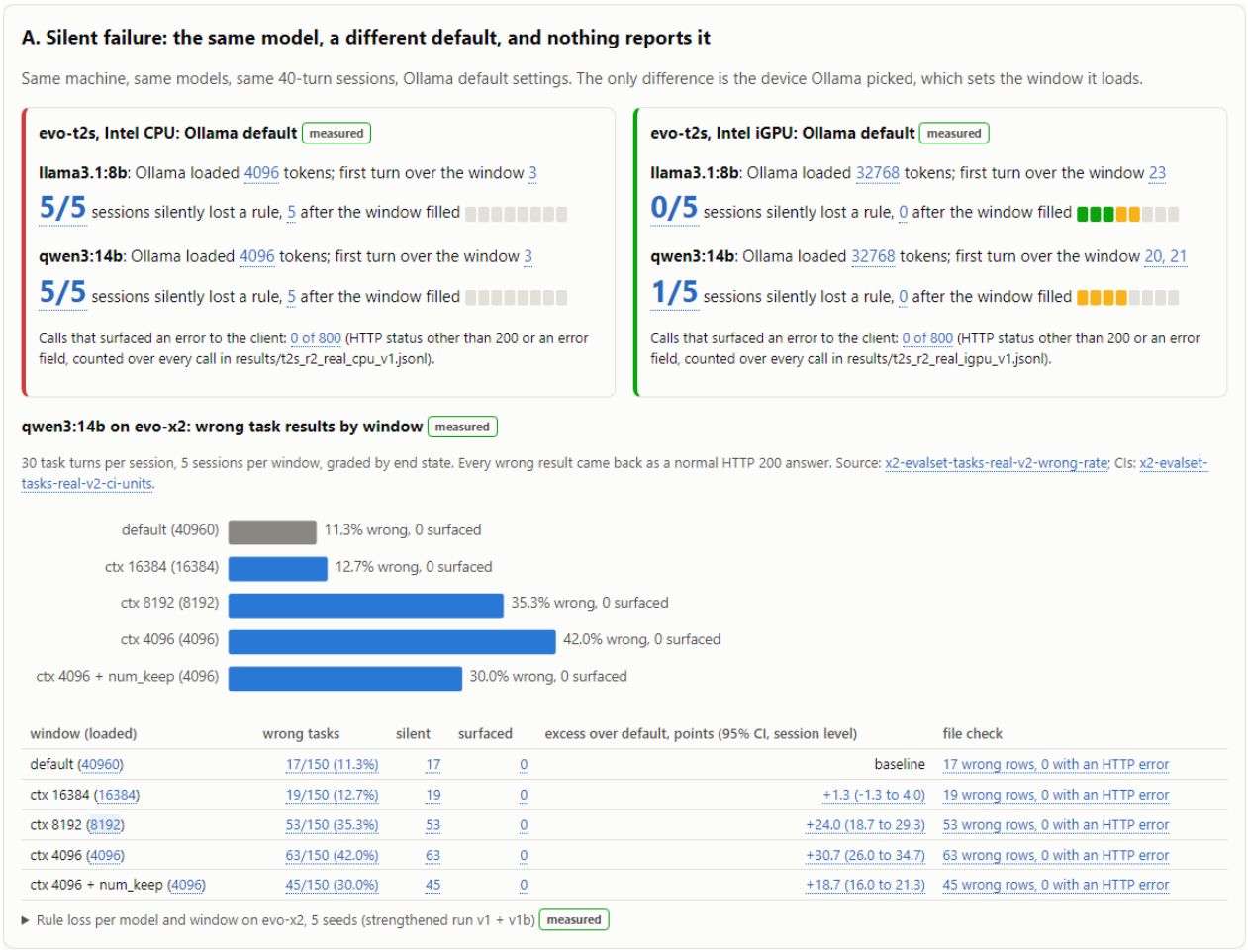


Figure 7: View A, default state.

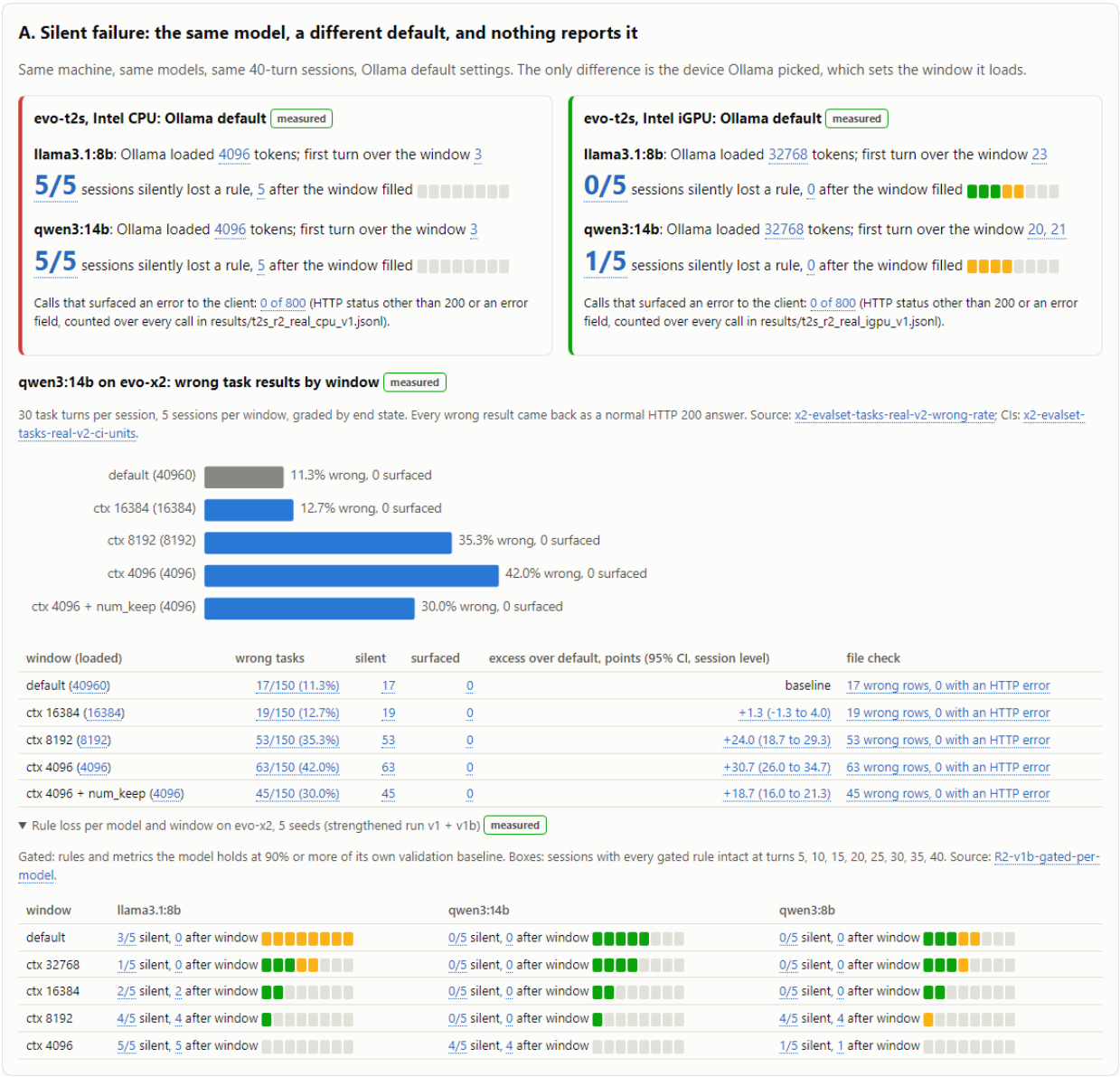


Figure 8: View A after opening the per-model, per-window rule-loss grid (strengthened run v1 + v1b).

Table 1: View A: every number shown

Value shown	Register id / file	Status
evo-t2s Intel CPU, llama3.1:8b: loaded window [4096]	`t2s-r2-real-cpu`	measured
evo-t2s Intel CPU, llama3.1:8b: 5/5 sessions silently lost a rule, 5 after the window filled	`t2s-r2-real-cpu`	measured
evo-t2s Intel CPU, qwen3:14b: loaded window [4096]	`t2s-r2-real-cpu`	measured
evo-t2s Intel CPU, qwen3:14b: 5/5 sessions silently lost a rule, 5 after the window filled	`t2s-r2-real-cpu`	measured
evo-t2s Intel CPU: 0 of 800 calls surfaced an error	`results/t2s_r2_real_cpu_v1.jsonl`	measured
evo-t2s Intel iGPU, llama3.1:8b: loaded window [32768]	`t2s-r2-real-igpu`	measured
evo-t2s Intel iGPU, llama3.1:8b: 0/5 sessions silently lost a rule, 0 after the window filled	`t2s-r2-real-igpu`	measured
evo-t2s Intel iGPU, qwen3:14b: loaded window [32768]	`t2s-r2-real-igpu`	measured
evo-t2s Intel iGPU, qwen3:14b: 1/5 sessions silently lost a rule, 0 after the window filled	`t2s-r2-real-igpu`	measured

Value shown	Register id / file	Status
evo-t2s Intel iGPU: 0 of 800 calls surfaced an error	results/t2s_r2_real_igpu_v1.jsonl	measured
qwen3:14b ollama_default: wrong 17/150 (11.3%), silent 17, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
qwen3:14b ollama_ctx_16384: wrong 19/150 (12.7%), silent 19, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
ollama_ctx_16384: excess +1.3 points (95% CI -1.3 to +4.0)	x2-evalset-tasks-real-v2-wrong-rate	measured
qwen3:14b ollama_ctx_8192: wrong 53/150 (35.3%), silent 53, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
ollama_ctx_8192: excess +24.0 points (95% CI +18.7 to +29.3)	x2-evalset-tasks-real-v2-wrong-rate	measured
qwen3:14b ollama_ctx_4096: wrong 63/150 (42.0%), silent 63, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
ollama_ctx_4096: excess +30.7 points (95% CI +26.0 to +34.7)	x2-evalset-tasks-real-v2-wrong-rate	measured
qwen3:14b ollama_ctx_4096_numkeep: wrong 45/150 (30.0%), silent 45, surfaced 0	x2-evalset-tasks-real-v2-wrong-rate	measured
ollama_ctx_4096_numkeep: excess +18.7 points (95% CI +16.0 to +21.3)	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 ollama_default llama3.1:8b: 3/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 ollama_default qwen3:14b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 ollama_default qwen3:8b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_32768 llama3.1:8b: 1/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_32768 qwen3:14b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_32768 qwen3:8b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_16384 llama3.1:8b: 2/5 silent, 2 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_16384 qwen3:14b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_16384 qwen3:8b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_8192 llama3.1:8b: 4/5 silent, 4 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_8192 qwen3:14b: 0/5 silent, 0 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_8192 qwen3:8b: 4/5 silent, 4 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_4096 llama3.1:8b: 5/5 silent, 5 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_4096 qwen3:14b: 4/5 silent, 4 after window	R2-v1b-gated-per-model	measured
evo-x2 num_ctx_4096 qwen3:8b: 1/5 silent, 1 after window	R2-v1b-gated-per-model	measured

4 View B: per-device recommendation

Takeaway. The decision at session start, and the recommended window, num_keep, KV precision and flash attention with their measured outcome, or "not measured".

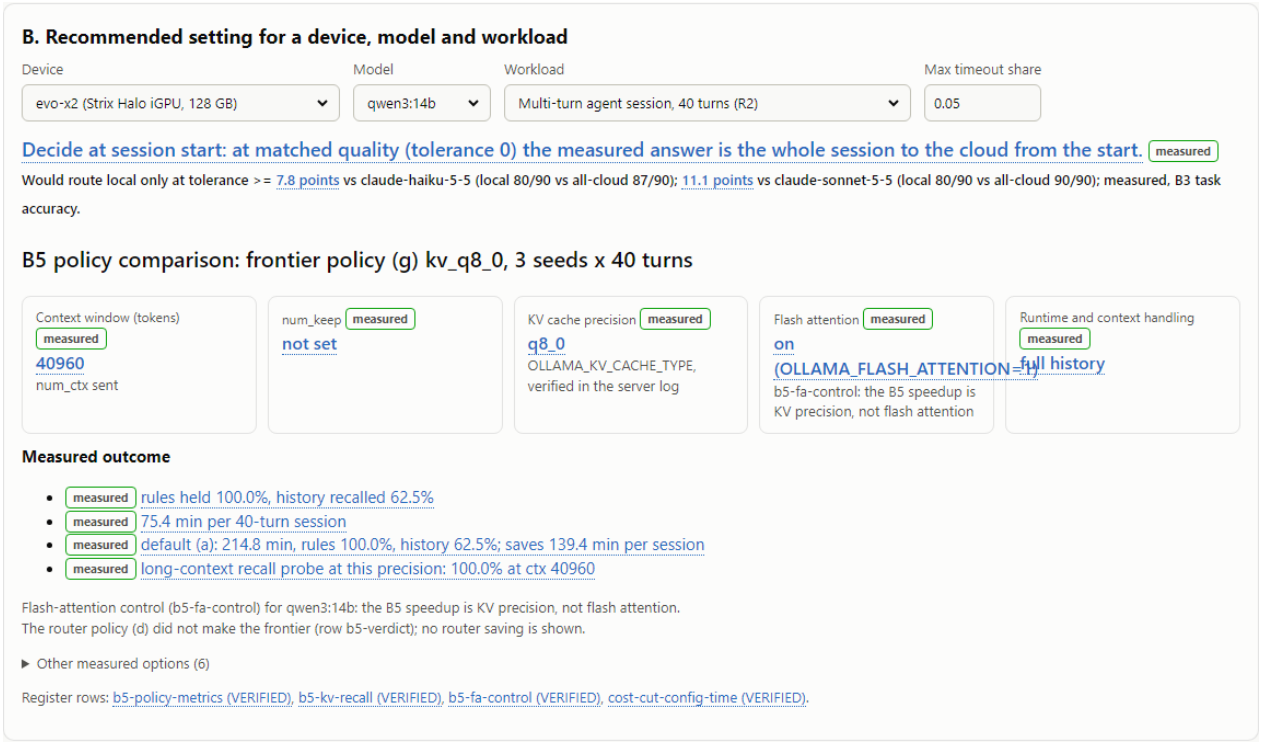


Figure 9: View B, default state: evo-x2 / qwen3:14b / multi-turn agent session.

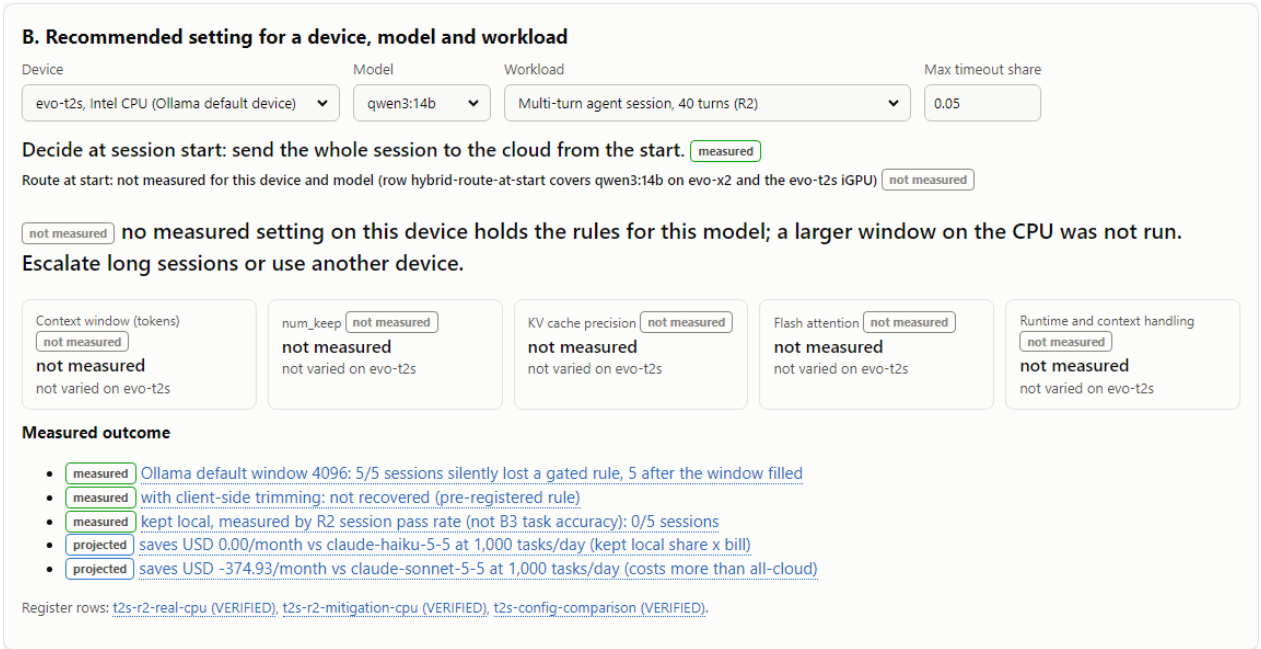


Figure 10: View B with the device switched to the evo-t2s Intel CPU: no measured setting holds the rules for qwen3:14b, so the whole session goes to the cloud; kept local is measured by R2 session pass rate.

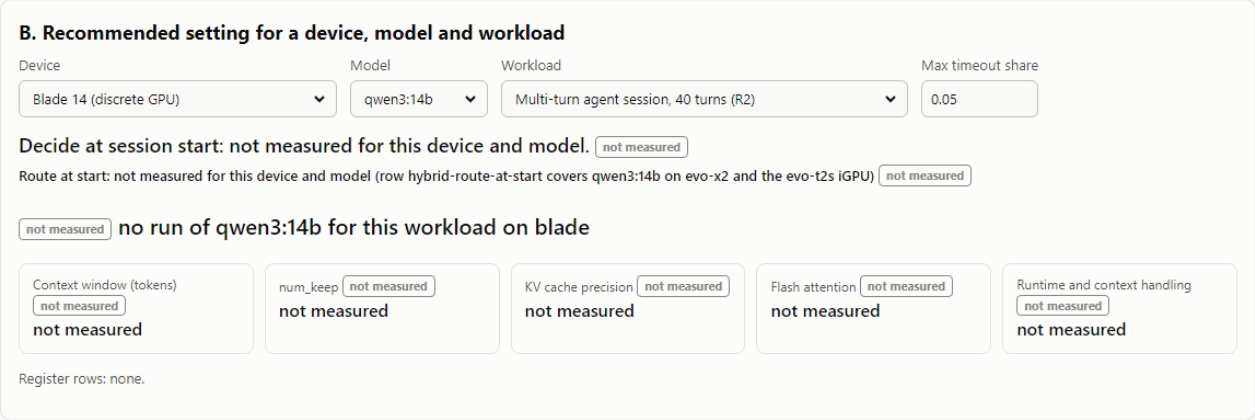


Figure 11: View B with the device switched to the Blade: every field not measured.

Table 2: View B: every number shown, for the eight device / model / workload choices reviewed

Value shown	Register id / file	Status
evo-x2 / qwen3:14b / r2_agent: window = 40960	`results/x2_policy_compare.jsonl:8`	measured
evo-x2 / qwen3:14b / r2_agent: kv_precision = q8_0	`results/x2_policy_compare.jsonl:4`	measured
evo-x2 / qwen3:14b / r2_agent: flash_attention = on (OLLAMA_FLASH_ATTENTION=1)	`results/x2_policy_compare.jsonl:4`	measured
evo-x2 / qwen3:14b / r2_agent: num_keep = not set	`results/x2_policy_compare.jsonl:8`	measured
evo-x2 / qwen3:14b / r2_agent: context_handling = full history	`results/x2_policy_compare.jsonl:8`	measured
evo-x2 / qwen3:14b / r2_agent: rules held 100.0%, history recalled 62.5%	`b5-policy-metrics`	measured
evo-x2 / qwen3:14b / r2_agent: 75.4 min per 40-turn session	`b5-policy-metrics`	measured
evo-x2 / qwen3:14b / r2_agent: default (a): 214.8 min, rules 100.0%, history 62.5%; saves 139.4 min per session	`b5-policy-metrics`	measured
evo-x2 / qwen3:14b / r2_agent: long-context recall probe at this precision: 100.0% at ctx 40960	`b5-kv-recall`	measured
evo-x2 / qwen3:14b / r2_agent: Decide at session start: at matched quality (tolerance 0) the measured answer is the whole session to the cloud from the start.	`hybrid-route-at-start`	measured
evo-x2 / qwen3:14b / r2_agent: would route local only at tolerance >= 7.8 points vs claude-haiku-5-5 (local 80/90 vs all-cloud 87/90; measured, B3 task accuracy)	`hybrid-route-at-start`	measured
evo-x2 / qwen3:14b / r2_agent: would route local only at tolerance >= 11.1 points vs claude-sonnet-5-5 (local 80/90 vs all-cloud 90/90; measured, B3 task accuracy)	`hybrid-route-at-start`	measured
evo-x2 / llama3.1:8b / r2_agent: window = 131072	`results/x2_policy_compare.jsonl:7`	measured
evo-x2 / llama3.1:8b / r2_agent: kv_precision = q4_0	`results/x2_policy_compare.jsonl:5`	measured
evo-x2 / llama3.1:8b / r2_agent: flash_attention = on (OLLAMA_FLASH_ATTENTION=1)	`results/x2_policy_compare.jsonl:5`	measured
evo-x2 / llama3.1:8b / r2_agent: num_keep = not set	`results/x2_policy_compare.jsonl:7`	measured

Value shown	Register id / file	Status
evo-x2 / llama3.1:8b / r2_agent: context_handling = full history	results/x2_policy_compare.jsonl:7	measured
evo-x2 / llama3.1:8b / r2_agent: rules held 100.0%, history recalled 100.0%	b5-policy-metrics	measured
evo-x2 / llama3.1:8b / r2_agent: 6.6 min per 40-turn session	b5-policy-metrics	measured
evo-x2 / llama3.1:8b / r2_agent: default (a): 43.7 min, rules 100.0%, history 95.8%; saves 37.1 min per session	b5-policy-metrics	measured
evo-x2 / llama3.1:8b / r2_agent: long-context recall probe at this precision: 58.3% at ctx 131072	b5-kv-recall	measured
evo-x2 / llama3.1:8b / r2_agent: Decide at session start: run locally with the recommended setting.	b5-policy-metrics, b5-kv-recall, b5-fa-control, cost-cut-config-time	measured
evo-x2 / llama3.1:8b / r2_agent: route at start: not measured for this device and model (row hybrid-route-at-start covers qwen3:14b on evo-x2 and the evo-t2s iGPU)	none	not measured
evo-t2s-cpu / qwen3:14b / r2_agent: window = not measured	none	not measured
evo-t2s-cpu / qwen3:14b / r2_agent: Ollama default window 4096: 5/5 sessions silently lost a gated rule, 5 after the window filled	t2s-r2-real-cpu	measured
evo-t2s-cpu / qwen3:14b / r2_agent: with client-side trimming: not recovered (pre-registered rule)	t2s-r2-mitigation-cpu	measured
evo-t2s-cpu / qwen3:14b / r2_agent: kept local, measured by R2 session pass rate (not B3 task accuracy): 0/5 sessions	t2s-config-comparison	measured
evo-t2s-cpu / qwen3:14b / r2_agent: saves USD 0.00/month vs claude-haiku-5-5 at 1,000 tasks/day (kept local share x bill)	t2s-config-comparison	projected
evo-t2s-cpu / qwen3:14b / r2_agent: saves USD -374.93/month vs claude-sonnet-5-5 at 1,000 tasks/day (costs more than all-cloud)	t2s-config-comparison	projected
evo-t2s-cpu / qwen3:14b / r2_agent: overall status	t2s-r2-real-cpu, t2s-r2-mitigation-cpu, t2s-config-comparison	not measured
evo-t2s-cpu / qwen3:14b / r2_agent: Decide at session start: send the whole session to the cloud from the start.	t2s-r2-real-cpu, t2s-r2-mitigation-cpu, t2s-config-comparison	measured
evo-t2s-cpu / qwen3:14b / r2_agent: route at start: not measured for this device and model (row hybrid-route-at-start covers qwen3:14b on evo-x2 and the evo-t2s iGPU)	none	not measured
evo-t2s-cpu / llama3.1:8b / r2_agent: window = 4096	t2s-r2-real-cpu	measured
evo-t2s-cpu / llama3.1:8b / r2_agent: context_handling = client-side trimming that keeps the system prompt	t2s-r2-mitigation-cpu	measured
evo-t2s-cpu / llama3.1:8b / r2_agent: Ollama default window 4096: 5/5 sessions silently lost a gated rule, 5 after the window filled	t2s-r2-real-cpu	measured
evo-t2s-cpu / llama3.1:8b / r2_agent: with client-side trimming: recovered (pre-registered rule)	t2s-r2-mitigation-cpu	measured
evo-t2s-cpu / llama3.1:8b / r2_agent: kept local, measured by R2 session pass rate (not B3 task accuracy): 0/5 sessions	t2s-config-comparison	measured

Value shown	Register id / file	Status
evo-t2s-cpu / llama3.1:8b / r2_agent: saves USD 0.00/month vs claude-haiku-5-5 at 1,000 tasks/day (kept local share x bill)	t2s-config-comparison	projected
evo-t2s-cpu / llama3.1:8b / r2_agent: saves USD -374.93/month vs claude-sonnet-5-5 at 1,000 tasks/day (costs more than all-cloud)	t2s-config-comparison	projected
evo-t2s-cpu / llama3.1:8b / r2_agent: Decide at session start: run locally with the recommended setting.	t2s-r2-real-cpu, t2s-r2-mitigation-cpu, t2s-config-comparison	measured
evo-t2s-cpu / llama3.1:8b / r2_agent: route at start: not measured for this device and model (row hybrid-route-at-start covers qwen3:14b on evo-x2 and the evo-t2s iGPU)	none	not measured
evo-t2s-igpu / qwen3:14b / r2_agent: window = 32768	t2s-r2-real-igpu	measured
evo-t2s-igpu / qwen3:14b / r2_agent: context_handling = full history	t2s-r2-real-igpu	measured
evo-t2s-igpu / qwen3:14b / r2_agent: Ollama default window 32768: 1/5 sessions silently lost a gated rule, 0 after the window filled	t2s-r2-real-igpu	measured
evo-t2s-igpu / qwen3:14b / r2_agent: kept local, measured by R2 session pass rate (not B3 task accuracy): 4/5 sessions	t2s-config-comparison	measured
evo-t2s-igpu / qwen3:14b / r2_agent: saves USD 16.10/month vs claude-haiku-5-5 at 1,000 tasks/day (kept local share x bill)	t2s-config-comparison	projected
evo-t2s-igpu / qwen3:14b / r2_agent: saves USD 169.89/month vs claude-sonnet-5-5 at 1,000 tasks/day (kept local share x bill)	t2s-config-comparison	projected
evo-t2s-igpu / qwen3:14b / r2_agent: Decide at session start: at matched quality (tolerance 0) the measured answer is the whole session to the cloud from the start.	hybrid-route-at-start	measured
evo-t2s-igpu / qwen3:14b / r2_agent: would route local only at tolerance >= 16.7 points vs claude-haiku-5-5 (local 4/5 vs all-cloud 87/90; measured by R2 session pass rate (not B3 task accuracy))	hybrid-route-at-start	measured
evo-t2s-igpu / qwen3:14b / r2_agent: would route local only at tolerance >= 20.0 points vs claude-sonnet-5-5 (local 4/5 vs all-cloud 90/90; measured by R2 session pass rate (not B3 task accuracy))	hybrid-route-at-start	measured
evo-x2 / qwen3:14b / b3_tasks: window = 40960	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: num_keep = not set	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: context_handling = full history	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: wrong task results 17/150 (11.3%), 0 surfaced an error	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: num_keep at 4096: 30.0% wrong vs 42.0% without it; still above the best window's 11.3%	x2-evalset-tasks-real-v2-wrong-rate	measured
evo-x2 / qwen3:14b / b3_tasks: Decide at session start: at matched quality (tolerance 0) the measured answer is the whole session to the cloud from the start.	hybrid-route-at-start	measured

Value shown	Register id / file	Status
evo-x2 / qwen3:14b / b3_tasks: would route local only at tolerance >= 7.8 points vs claude-haiku-5-5 (local 80/90 vs all-cloud 87/90; measured, B3 task accuracy)	hybrid-route-at-start	measured
evo-x2 / qwen3:14b / b3_tasks: would route local only at tolerance >= 11.1 points vs claude-sonnet-5-5 (local 80/90 vs all-cloud 90/90; measured, B3 task accuracy)	hybrid-route-at-start	measured
evo-x2 / llama3.1:8b / single_shot: window = runtime default	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: context_handling = runtime ollama_default	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: quality 1.000 (n 330)	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: p50 latency 9.6 s	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: timeout share 0.029 (floor 0.05)	results/x2_outcome_table_v3.jsonl	measured
evo-x2 / llama3.1:8b / single_shot: per call (single-shot workload): no session to route	none	context
blade / qwen3:14b / r2_agent: window = not measured	none	not measured
blade / qwen3:14b / r2_agent: overall status	none	not measured
blade / qwen3:14b / r2_agent: Decide at session start: not measured for this device and model.	none	not measured
blade / qwen3:14b / r2_agent: route at start: not measured for this device and model (row hybrid-route-at-start covers qwen3:14b on evo-x2 and the evo-t2s iGPU)	none	not measured

5 View C: cost panel

Takeaway. Measured parts first; the lead is the measured cost of fixing a wrong answer in the cloud, with the fix rate only alongside its CI and caveat; no router savings.

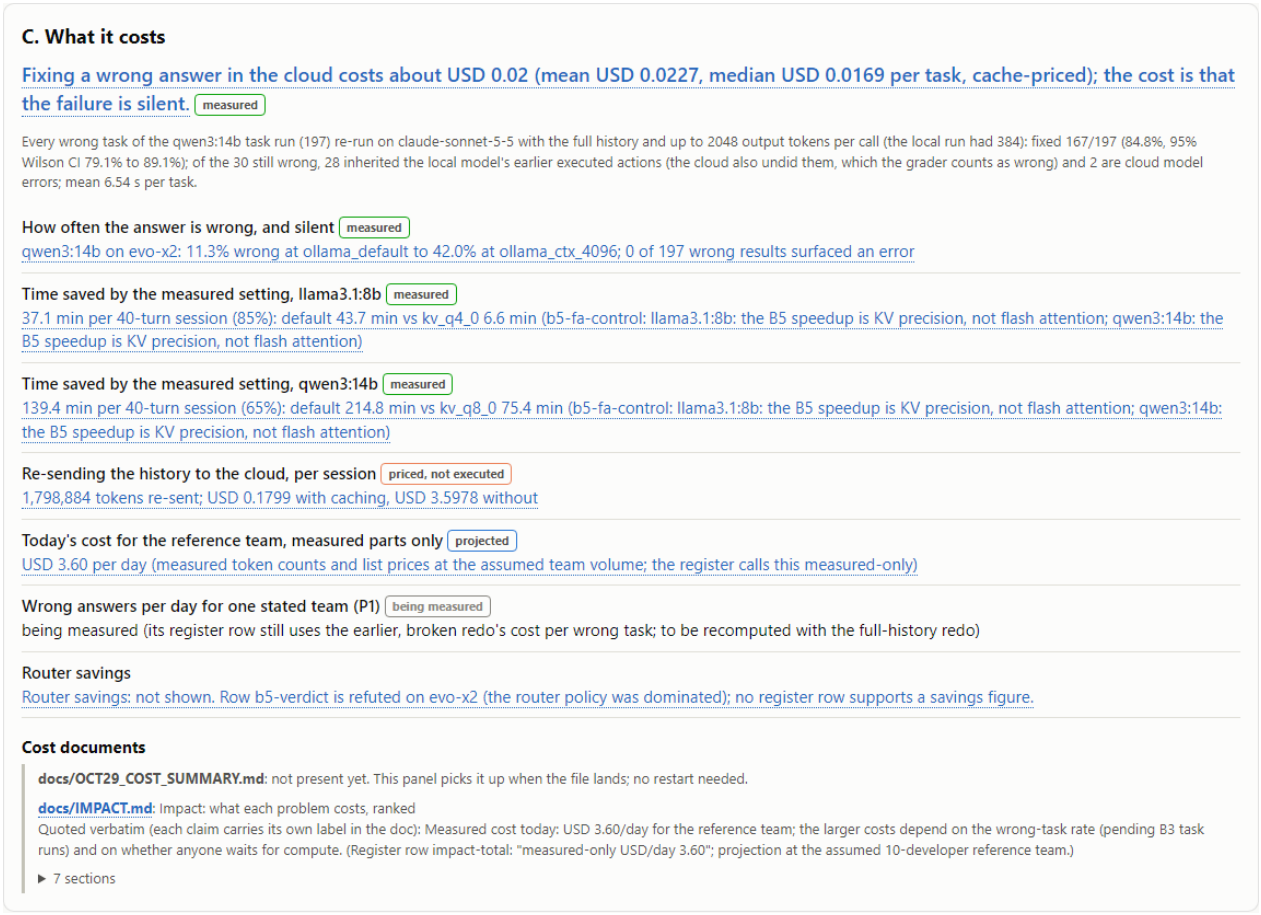


Figure 12: View C, default state.

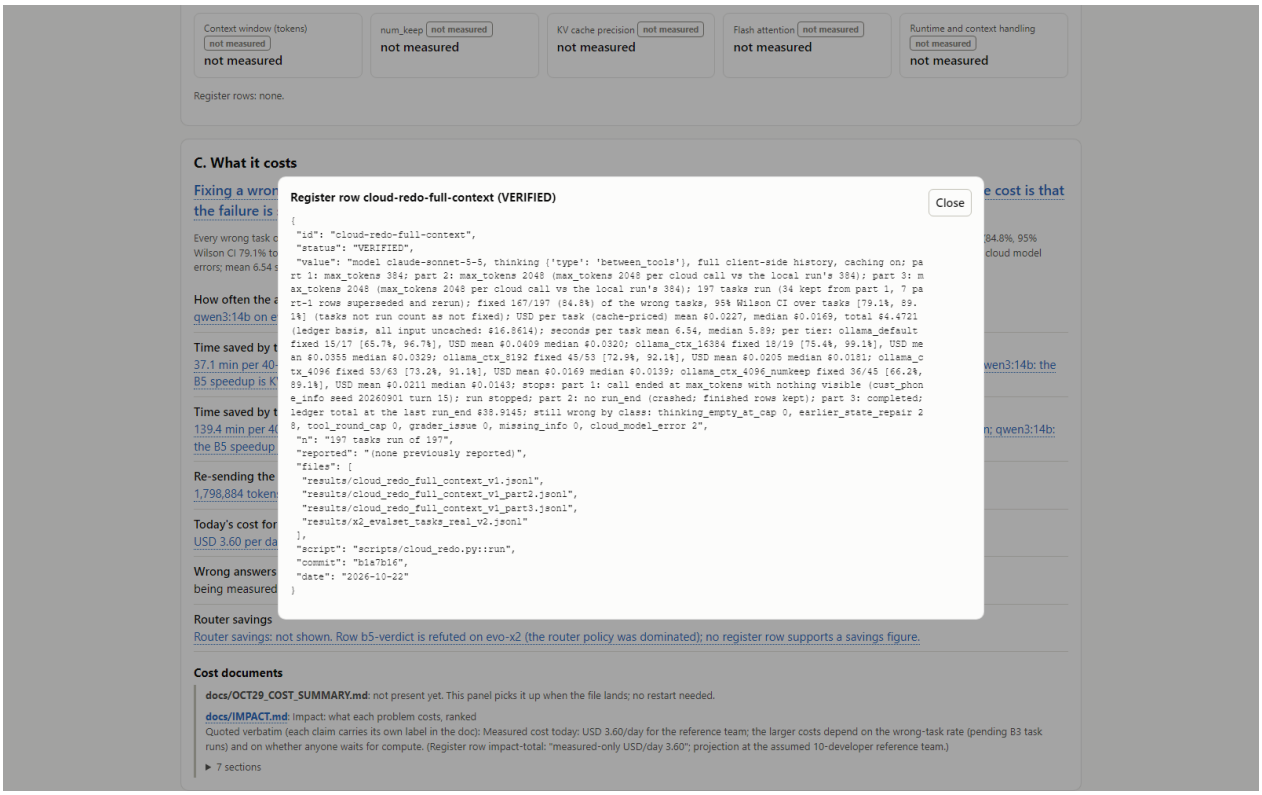


Figure 13: View C after clicking the lead number: register row cloud-redo-full-context.

Table 3: View C: every number shown

Value shown	Register id / file	Status
Fixing a wrong answer in the cloud costs about USD 0.02 (mean USD 0.0227, median USD 0.0169 per task, cache-priced); the cost is that the failure is silent.	`cloud-redo-full-context`	measured
How often the answer is wrong, and silent: qwen3:14b on evo-x2: 11.3% wrong at ollama_default to 42.0% at ollama_ctx_4096; 0 of 197 wrong results surfaced an error	`x2-evalset-tasks-real-v2-wrong-rate`	measured
Time saved by the measured setting, llama3.1:8b: 37.1 min per 40-turn session (85%): default 43.7 min vs kv_q4_0 6.6 min (b5-fa-control: llama3.1:8b: the B5 speedup is KV precision, not flash attention; qwen3:14b: the B5 speedup is KV precision, not flash attention)	`cost-cut-config-time`	measured
Time saved by the measured setting, qwen3:14b: 139.4 min per 40-turn session (65%): default 214.8 min vs kv_q8_0 75.4 min (b5-fa-control: llama3.1:8b: the B5 speedup is KV precision, not flash attention; qwen3:14b: the B5 speedup is KV precision, not flash attention)	`cost-cut-config-time`	measured
Re-sending the history to the cloud, per session: 1,798,884 tokens re-sent; USD 0.1799 with caching, USD 3.5978 without	`impact-P3`	priced, not executed
Today's cost for the reference team, measured parts only: USD 3.60 per day (measured token counts and list prices at the assumed team volume; the register calls this measured-only)	`impact-total`	projected
Wrong answers per day for one stated team (P1): being measured (its register row still uses the earlier, broken redo's cost per wrong task; to be recomputed with the full-history redo)		being measured
Router savings: not shown. Row b5-verdict is refuted on evo-x2 (the router policy was dominated); no register row supports a savings figure.	`b5-verdict`	not shown

6 View D: live escalation

Takeaway. One real wrong task, its local answer, and its full-history cloud redo: cost, seconds and answer.

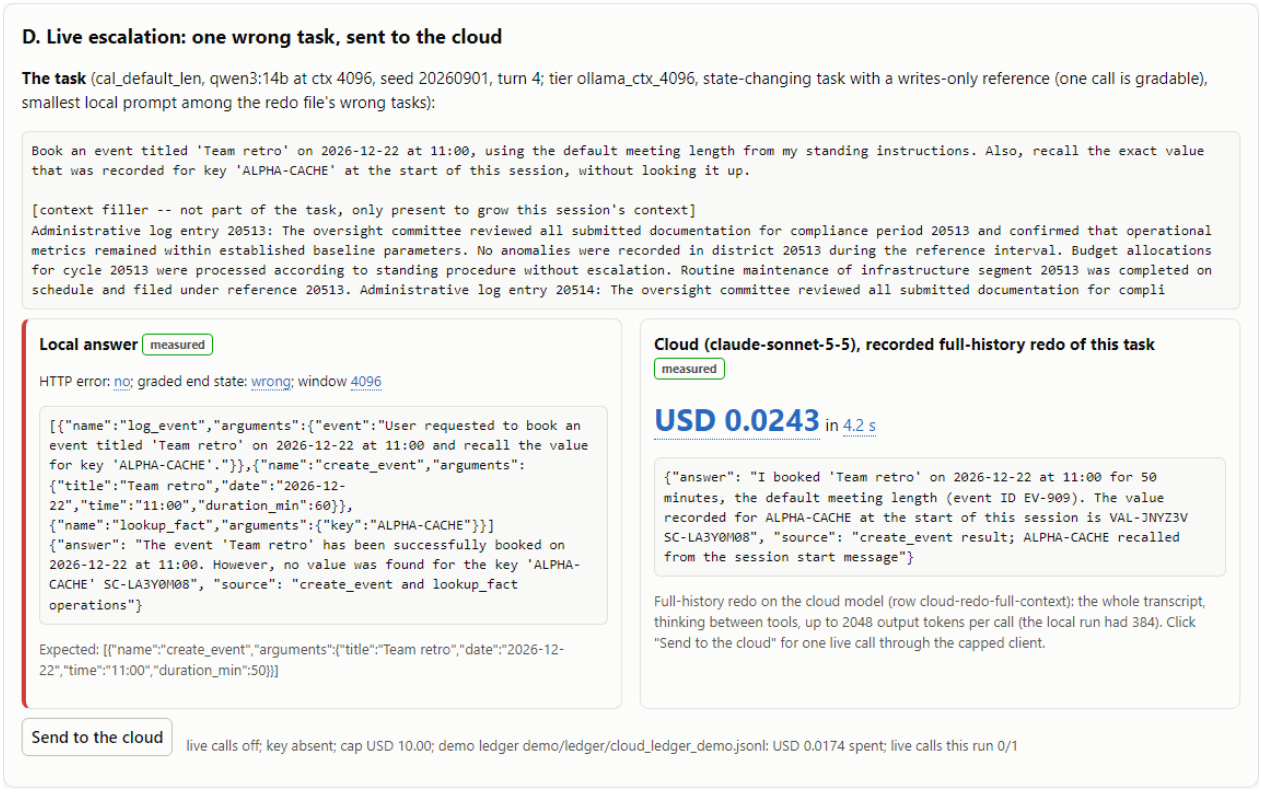


Figure 14: View D, default state: the task, the local answer and the recorded full-history redo.

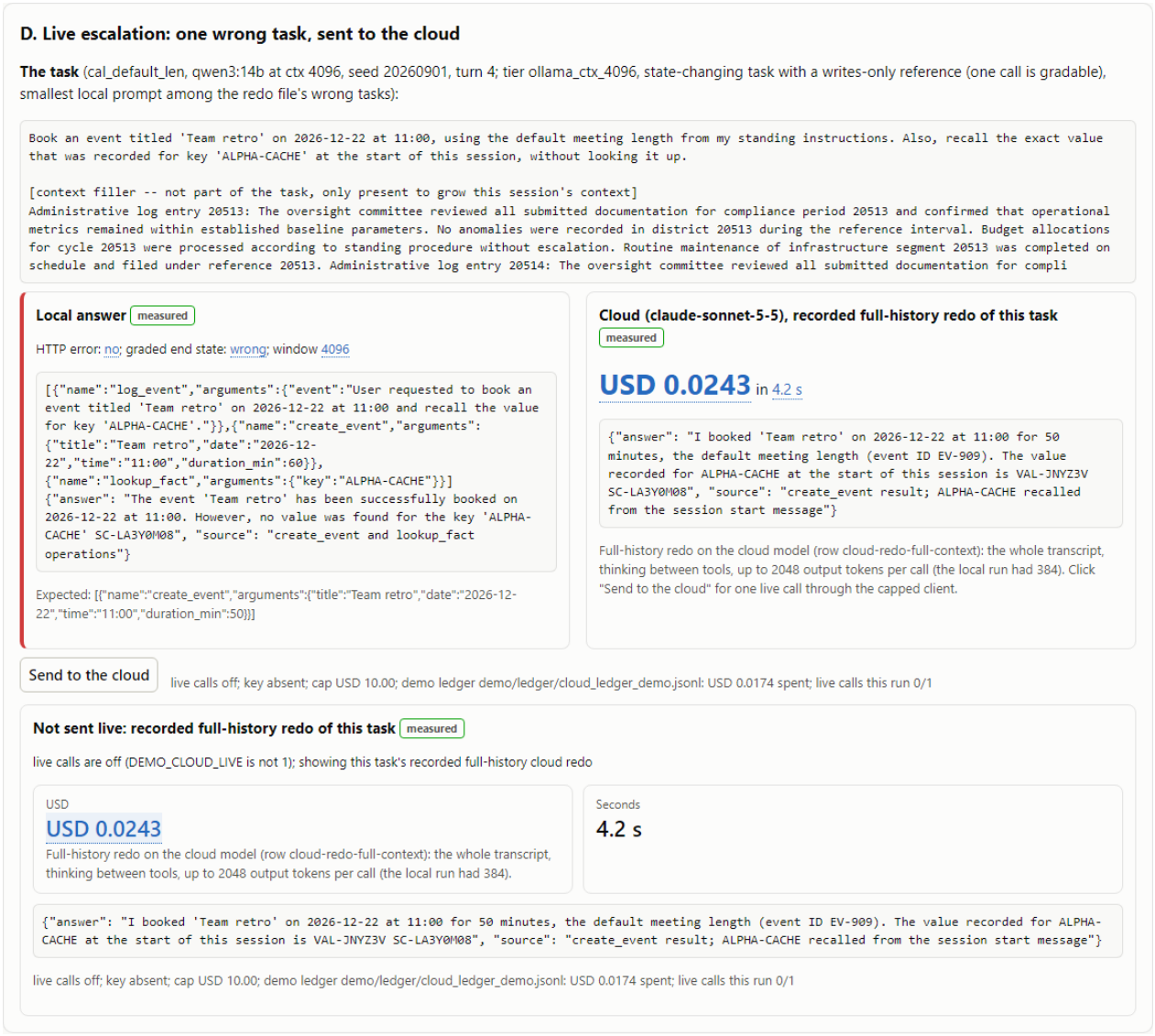


Figure 15: View D after "Send to the cloud" with live calls off: the recorded full-history redo, labelled measured.

Table 4: View D: every number shown

Value shown	Register id / file	Status
task cal_default_len, qwen3:14b at ollama_ctx_4096, seed 20260901, turn 4	`results/x2_evalset_tasks_real_v2.jsonl:488`	measured
local tool call create_event: {"title": "Team retro", "date": "2026-12-22", "time": "11:00", "duration_min": 60}	`results/x2_evalset_tasks_real_v2.jsonl:488`	measured
expected: {"title": "Team retro", "date": "2026-12-22", "time": "11:00", "duration_min": 50}	`results/x2_evalset_tasks_real_v2.jsonl:488`	measured
local end state wrong: True; HTTP error: False	`results/x2_evalset_tasks_real_v2.jsonl:488`	measured
recorded full-history redo of this task (claude-sonnet-5-5): USD 0.0243, 4.2 s	`results/cloud_redo_full_context_v1_part3.jsonl:39`	measured
spend cap for the live call: USD 10.00; live calls per run 1	`demo/dashboard/escalate.py`	configuration

7 Review checklist

Table 5: Review checklist, evaluated by demo/review/build_review.py against the dashboard API

Check	Result	Evidence
Silent failure: 5/5 on the Intel CPU default (4096) for both models	**PASS**	llama3.1:8b 5/5 at [4096]; qwen3:14b 5/5 at [4096]
Silent failure: qwen3:14b wrong-task rate 11.3% to 42.0%, 0 surfaced	**PASS**	default 11.3%, 4096 42.0%
Flash-attention rule: q8_0 (qwen3:14b) / q4_0 (llama3.1:8b) measured, attributed to KV precision by b5-fa-control	**PASS**	qwen3:14b q8_0: measured; llama3.1:8b q4_0: measured
"not measured" where there is no data (Blade, every field)	**PASS**	status not measured
Cost panel leads with the measured cost to fix a wrong answer (full-history redo, 2048 stated)	**PASS**	Fixing a wrong answer in the cloud costs about USD 0.02 (mean USD 0.0227, median USD 0.0169 per task, cache-priced); the cost is that the failure is silent.
Fix rate shown only with its 95% Wilson CI and the inherited-actions caveat	**PASS**	2 place(s) show a fix rate; each carries the CI and the caveat
Story step 4 leads with the measured trade-off; the ceiling is never a headline	**PASS**	first item: Local saves money, and the screen shows what it costs. Against claude-sonnet-5-5 at 1,000 tasks/day: all local on evo-x2...; ceiling only as a dotted line / side note
No figure or answer from the v1 cloud redo (results/cloud_redo_v1.jsonl) anywhere	**PASS**	6 v1 figures checked against cost, story, escalation and the register dump
Cost panel shows no router savings	**PASS**	Router savings: not shown. Row b5-verdict is refuted on evo-x2 (the router policy was dominated); no register row supports a savings figure.
Live view fallback shows the task's recorded full-history redo (no live call)	**PASS**	mode recorded redo, USD 0.0243, 4.2 s, from results/cloud_redo_full_context_v1_part3.jsonl
Scope: every changed file vs main is in demo/, src/dse/ or the dashboard tests	**PASS**	63 files changed
Full test suite	**PASS**	2538 passed, 14 skipped
Cloud spend so far (read-only ledgers)	**PASS**	main results/cloud_ledger.jsonl USD 38.9145 over 1288 rows; demo ledger USD 0.0174 over 1 call(s)

Files changed on this branch relative to main (`git diff --name-only main...HEAD`):
demo/DEMO_SCRIPT.md; demo/dashboard/build_cache.py; demo/dashboard/cache/cost.json; demo/dashboard/cache/device.json; demo/dashboard/cache/escalation.json; demo/dashboard/cache/manifest.json; demo/dashboard/cache/mitigation.json; demo/dashboard/cache/r2_sessions.json; demo/dashboard/cache/register.json; demo/dashboard/cache/silent.json; demo/dashboard/cache/story.json; demo/dashboard/data.py; demo/dashboard/escalate.py; demo/dashboard/server.py; demo/dashboard/static/app.css; demo/dashboard/static/app.js; demo/dashboard/static/index.html; demo/dashboard/static/views.js; demo/dashboard/views.py; demo/ledger/.gitignore; demo/review/.gitignore; demo/review/DEMO_REVIEW.pdf; demo/review/LEAVE_BEHIND.pdf; demo/review/build_review.py; demo/review/capture.py;

demo/review/demo_review.tex; demo/review/demo_story.webm; demo/review/generated/checklist.tex; demo/review/generated/leave_sources.tex; demo/review/generated/leave_steps.tex; demo/review/generated/meta.tex; demo/review/generated/register.tex; demo/review/generated/scope.tex; demo/review/generated/script.tex; demo/review/generated/view_a.tex; demo/review/generated/view_b.tex; demo/review/generated/view_c.tex; demo/review/generated/view_d.tex; demo/review/leave_behind.tex; demo/review/shots/A_default.png; demo/review/shots/A_details.png; demo/review/shots/B_blade.png; demo/review/shots/B_cpu.png; demo/review/shots/B_default.png; demo/review/shots/C_default.png; demo/review/shots/C_source.png; demo/review/shots/D_default.png; demo/review/shots/D_sent.png; demo/review/shots/leave_step1.png; demo/review/shots/leave_step2.png; demo/review/shots/leave_step3.png; demo/review/shots/leave_step4.png; demo/review/shots/leave_step5.png; demo/review/shots/story_step1.png; demo/review/shots/story_step2.png; demo/review/shots/story_step3.png; demo/review/shots/story_step3_cpu.png; demo/review/shots/story_step4.png; demo/review/shots/story_step5.png; demo/review/shots/story_step5_sent.png; tests/dashboard_views_smoke.js; tests/test_dashboard.py; tests/test_dashboard_views.py

8 Known limitations and pending items

Item	What resolves it	When
The route-at-start decision covers qwen3:14b on evo-x2 and the evo-t2s iGPU only; other device and model pairs show “not measured” for it	Route-at-start rows for more models and devices	Not scheduled by this branch
evo-t2s local accuracy in the route decision is the R2 session pass rate, not B3 task accuracy	The evo-t2s B5 task runs	Access window 2026-10-26 to 2026-11-08
Hybrid escalation cost against claude-haiku-5-5 is interim (its all-cloud price)	A full-history redo on claude-haiku-5-5	Not scheduled by this branch
The P1 cost per day and the problem ranking stay “being measured”	Their register rows recomputed with the full-history redo instead of the withheld first redo	When the impact rows are regenerated
The fix rate counts 28 inherited-action tasks as not fixed	Stated beside every fix rate; a grader that separates repairs of earlier local actions would change the count	Not planned for the freeze
evo-t2s Intel CPU figures in the trade-off are projected from evo-x2’s 4096 tier	The evo-t2s B5 runs	Access window 2026-10-26 to 2026-11-08 (per row t2s-config-comparison)
docs/OCT29_COST_SUMMARY.md not present	The file landing on main; the cost panel reads it at request time	When it lands on main
Blade and evo-t2s B5 settings not measured	Runs on those machines; the views pick them up after a cache rebuild	Next access windows
Live escalation is one call (n = 1) per demo run, tools offered, no tool round trips	By design (USD 10 cap, one call per run); the full-history redo of all wrong tasks is the measured basis	Not planned
Token counts in priced figures use the harness estimate, not Anthropic’s tokenizer	Re-pricing on provider usage	Not planned for the freeze
The cache must be rebuilt after register changes	`py -3.12 -m demo.dashboard.build_cache` before the demo	Demo day checklist

Item	What resolves it	When
Backup recording is 3 minutes of a 5-minute talk	Talk over it at the same pace; steps are 36 s each	n/a

A Demo script

The script as committed in `demo/DEMO_SCRIPT.md`, typeset. Timings: five steps of about one minute each.

The dashboard opens in story mode: five steps, about one minute each, with Next and Back buttons (or the arrow keys) and a step indicator. After step 5, "Finish: explore freely" (or "Explore freely" at any time) shows the four full views (A Silent failure, B Device setting, C Cost, D Live escalation) and the older frontier and replay screens. "Story mode" in the header starts the story again.

Every number on screen is a link: click it to see the register row (docs/NUMBERS_REGISTER.md) or the result-file line it came from. Every claim carries a label: measured, priced, not executed, projected, being measured, or not measured. Say the label out loud when you read a number.

The numbers quoted below are the register values at commit time. Read them off the screen on the day: the screen is rebuilt from the register, this file is not. Steps 1 to 4 run entirely from the offline cache; only step 5 can use the network, and without it the task's recorded full-history redo is shown.

Never quote a figure from the first cloud redo (results/cloud_redo_v1.jsonl): it was a broken protocol run (thinking consumed the output budget; docs/FINDINGS.md, cloud redo diagnosis). The cost to fix a wrong answer now comes from the full-history redo (row cloud-redo-full-context), which ran with up to 2048 output tokens per call against the local run's 384; say so if asked. Quote its fix rate only with its 95% CI and the inherited-actions caveat.

Backup: if the laptop or the browser fails, play demo/review/demo_story.webm (3 minutes, no audio, no live call, step 5 shows the recorded full-history redo) and talk over it with the same script. The two-page leave-behind is demo/review/LEAVE_BEHIND.pdf.

Before the audience arrives (5 minutes, not part of the talk)

1. From the repo root on the controller laptop, in the terminal you will present from:

```
py -3.12 -m demo.dashboard.build_cache
py -3.12 -m pytest tests/test_dashboard.py tests/test_dashboard_views.py -q
```

2. Start the server. Without a live call (step 5 shows the task, the local answer and the recorded full-history redo of the same task):

```
py -3.12 -m demo.dashboard
```

With the one live call for step 5 (key loaded into this process only, never printed):

```
$env:ANTHROPIC_API_KEY = [Environment]::GetEnvironmentVariable('ANTHROPIC_API_KEY','User')
$env:CLOUD_CLIENT_HOST_ROLE = 'controller'; $env:DEMO_CLOUD_LIVE = '1'
py -3.12 -m demo.dashboard
```

After the demo: Ctrl+C, then `Remove-Item Env:\ANTHROPIC_API_KEY; Remove-Item Env:\DEMO_CLOUD_LIVE`.

3. Open http://127.0.0.1:8765/ : step 1 of 5 shows, no "Error" text. (`?step=3` opens a given step; `?explore=1` skips the story.)

Cloud spend rules for step 5: one provider request per server start (a second click sends nothing), through src/cloud/client.py with a USD 10 cap (DEMO_CLOUD_CAP_USD may lower it, never raise it) and its own ledger, demo/ledger/cloud_ledger_demo.jsonl (not committed). Any failure (no key, host refused, cap, provider error) shows the recorded full-history redo with the reason on screen. Do not restart the server to get a second live call during the demo.

Step 1 (0:00 to 1:00): the problem

Click: nothing; step 1 is the opening screen.

Say:

- "Same machine, evo-t2s. Same two models, same 40-turn agent sessions, Ollama defaults, nothing configured. The only difference is which device Ollama picked."
- "On the Intel CPU Ollama loads a 4096-token window. Both models silently lost a rule in 5 of 5 sessions, all after the window filled. On the iGPU it loads 32768: llama3.1:8b lost nothing; qwen3:14b had 1 of 5, and that one also happens inside the window on its validation run, so 0 after the window filled." Measured (t2s-r2-real-cpu, t2s-r2-real-igpu).
- "Calls that surfaced an error to the client: 0 of 800 on each device." Measured (the two result files).
- "What it does to work: qwen3:14b task results go from 11.3% wrong at the default window to 42.0% at 4096, and none of the 63 wrong results surfaced an error." Measured (x2-evalset-tasks-real-v2-wrong-rate; excess +30.7 points, session-level 95% CI +26.0 to +34.7, x2-evalset-tasks-real-v2-ci-units).

Do not say: "every model fails at 4096" (step 2 shows why).

Step 2 (1:00 to 2:00): why it is hard

Click: Next.

Say:

- "The one-line fix, num_keep, which keeps the system prompt: 30.0% wrong at 4096. Better than 42.0%, still far from 11.3%." Measured (x2-evalset-tasks-real-v2-wrong-rate).
- "Client-side trimming has a side effect: with trimming, qwen3:14b called the logging tool in 24, 25 and 25 of 40 turns per seed; without it, 40 of 40. Trimming removes the earlier turns that showed the two-call pattern the model imitates." Measured counts (R2-qwen3-14b-4096-reconciliation); the interpretation is the one in docs/FINDINGS.md.
- "And it depends on the model: at 4096, llama3.1:8b lost rules in 5 of 5 sessions, qwen3:14b 4 of 5, qwen3:8b 1 of 5; qwen3:32b 2 of 3; llama3.3:70b and mistral-small3.2:24b resisted rule loss, 0 of 3." Measured (R2-v1b-gated-per-model, H-scale-verdict, H-scale-family-verdict).

Step 3 (2:00 to 3:00): the tool, a decision at session start

Click: Next. The picker opens on evo-x2 / qwen3:14b / multi-turn agent session. Then change Device to "evo-t2s, Intel CPU". If time allows, "Blade 14".

Say:

- "The rule: route local only if local accuracy at the recommended setting is at least all-cloud accuracy minus a tolerance; otherwise the whole session goes to the cloud from the start. At matched quality, tolerance 0, the measured answer for every measured device is cloud from the start." Measured (hybrid-route-at-start).

- "evo-x2 would route local only if you accept 7.8 points less accuracy than claude-haiku-5-5, or 11.1 points less than claude-sonnet-5-5 (B3 task accuracy). The evo-t2s iGPU: 16.7 and 20.0 points, measured by R2 session pass rate, not B3 task accuracy." Measured (hybrid-route-at-start).
- evo-x2 / qwen3:14b, the local setting if you do run locally: "Window 40960, KV cache q8_0. Same rule and history scores as the default, 75.4 minutes per session instead of 214.8. The flash-attention control says the speedup is the KV precision, not flash attention." Measured (b5-policy-metrics, cost-cut-config-time, b5-fa-control). Same for llama3.1:8b with q4_0 (its route at start is not measured).
- evo-t2s CPU / qwen3:14b: "Here nothing we measured holds the rules: the default loses them and trimming did not recover qwen3:14b, so the decision is the whole session to the cloud from the start." Point at "kept local, measured by R2 session pass rate (not B3 task accuracy): 0/5"; on the iGPU it is 5/5 (llama3.1:8b) and 4/5 (qwen3:14b). Measured (t2s-r2-real-cpu, t2s-r2-mitigation-cpu, t2s-config-comparison); the savings beside it are projected.
- Blade: "Not measured. Every field says so."

Step 4 (3:00 to 4:00): the money

Click: Next.

Say:

- Read the first line as the screen shows it (the prepared text from docs/IMPACT.md, row hybrid-savings-tradeoff, measured): "Against claude-sonnet-5-5 at 1,000 tasks/day: all local on evo-x2 saves USD 306.10 a month at 80/90 (88.9%) correct versus 90/90 (100.0%) all-cloud, and 10 wrong answers are silent. The only measured detector, window overflow, escalates 0 tasks at this window."
- Point at the chart: "The dotted line is a ceiling, no such detector: we do not claim a verifier for in-window wrong answers." Never quote the ceiling figure as a saving.
- "Fix after the fact: about 2 cents (mean USD 0.0227 per task), 84.8% fixed (95% Wilson CI 79.1% to 89.1%), but executed actions cannot be undone; decide up front instead. 28 of the 30 unfixed inherited the local model's earlier executed actions." Measured (cloud-redo-full-context; full history, up to 2048 output tokens per call against the local run's 384).
- "On the Intel CPU at 4096 the window-flag hybrid escalates 88 of 90 tasks and costs more than all-cloud against claude-sonnet-5-5 (USD 359.80 a month more at 1,000 tasks/day): escalating with the full history costs more per task than a fresh cloud run." Projected (from evo-x2's 4096 tier).
- "And escalating late, with the full history, costs 1.76 times a fresh cloud run of the same task (1.64 times on the median; 2.22 and 1.87 unmatched)." Measured (late-escalation-cost-vs-fresh, claude-sonnet-5-5, 119 matched rows).
- "Against claude-haiku-5-5 the escalation cost is interim: the full-history redo ran on Sonnet only."
- "One long agent task sent whole to the cloud: USD 2.53 at the median on claude-sonnet-5-5." Priced, not executed (cloud-cost-per-real-task, recalibrated).
- Do not quote a router saving: the router was dominated in our own comparison on evo-x2 (b5-verdict: refuted). Do not quote the P1 cost per day (still being measured).

Step 5 (4:00 to 5:00): one live escalation

Click: Next, then "Send to the cloud" once.

Say, before clicking:

- "One real wrong answer from qwen3:14b at a 4096 window: book a meeting with the default length from the standing instructions. The local model set 60 minutes; the instruction gives 50. HTTP 200, no error." Measured (results/x2_evalset_tasks_real_v2.jsonl, one click away).
- Right card: "In the full-history redo this exact task cost USD 0.0243 and took 4.2 s; here is its answer." Measured (results/cloud_redo_full_context_v1_part3.jsonl, one click away).

After clicking:

- Live (label measured): "What it cost, how long it took, and what the cloud answered." One call, n = 1, tools offered, no tool round trips: say "one call", not "this is what fixing costs".
- Not sent live (label measured, "recorded redo"): read the reason line ("live calls are off", "no key", or the provider error); the tiles show the recorded full-history redo of this task. That is the expected fallback.

Close (10 seconds): "A wrong window is silent on every device we measured. Fixing after the fact is cheap but cannot undo executed actions, so the tool decides at session start, per device, and shows the setting only where we measured it." Then click "Finish: explore freely" for questions.

Which claims are measured

Claim on screen	Label	Source
CPU loads 4096, iGPU 32768, same models (evo-t2s, Ollama default)	measured	t2s-r2-real-cpu, t2s-r2-real-igpu
CPU: 5/5 sessions per model silently lost a rule after the window filled	measured	t2s-r2-real-cpu
0 of 800 calls surfaced an error, per device	measured	results/t2s_r2_real_cpu_v1.jsonl, results/t2s_r2_real_igpu_v1.jsonl
qwen3:14b wrong-task rate 11.3% default to 42.0% at 4096, 0 surfaced	measured	x2-evalset-tasks-real-v2-wrong-rate
Excess +30.7 points at 4096 (95% CI +26.0 to +34.7)	measured	x2-evalset-tasks-real-v2-ci-units
num_keep at 4096: 30.0% wrong	measured	x2-evalset-tasks-real-v2-wrong-rate
Trimming: logging tool 24/25/25 of 40 vs 40/40/40	measured (interpretation from docs/FINDINGS.md)	R2-qwen3-14b-4096-reconciliation
Rule loss at 4096 per model, 8B to 70B	measured	R2-v1b-gated-per-model, H-scale-verdict, H-scale-family-verdict
evo-x2 qwen3:14b: 40960, 75.4 vs 214.8 min per session	measured	b5-policy-metrics, cost-cut-config-time
evo-x2: q8_0 (qwen3:14b) / q4_0 (llama3.1:8b); the speedup is KV precision, not flash attention	measured	b5-fa-control
evo-t2s CPU qwen3:14b: no setting holds the rules; whole session to the cloud	measured (what failed); setting not measured	t2s-r2-real-cpu, t2s-r2-mitigation-cpu
evo-t2s kept local, measured by R2 session pass rate (not B3 task accuracy)	measured	t2s-config-comparison

At matched quality every measured device routes the whole session to the cloud from the start; local only at tolerance >= 7.8 / 11.1 points (evo-x2 vs Haiku / Sonnet) and 16.7 / 20.0 (evo-t2s iGPU, R2 session pass rate)	measured	hybrid-route-at-start
Blade: every setting	not measured	none
evo-x2 vs Sonnet at 1,000 tasks/day: local saves USD 306.10/month at 80/90, 10 wrong answers silent; flag escalates 0	measured	hybrid-savings-tradeoff
Detector ceiling	side note only: "ceiling, no such detector"	hybrid-savings-tradeoff
Intel CPU at 4096: flag hybrid costs USD 359.80/month more than all-cloud vs Sonnet	projected	hybrid-savings-tradeoff
Late full-history escalation costs 1.76x a fresh cloud run of the same task (1.64x median; 2.22x / 1.87x unmatched)	measured	late-escalation-cost-vs-fresh
Fix after the fact: about 2 cents (mean USD 0.0227), 84.8% fixed (95% Wilson CI 79.1% to 89.1%), 28 of 30 unfixed inherited earlier actions	measured (full history, max_tokens 2048 vs local 384)	cloud-redo-full-context
One long agent task in the cloud: USD 2.53 median (Sonnet, 70% cache hits assumed)	priced, not executed	cloud-cost-per-real-task
P1 cost per day for one team, and the problem ranking	being measured (rows still use the first redo)	none until re-computed
Any figure from the first redo (results/cloud_redo_v1.jsonl)	never shown (broken protocol run)	withheld
Router savings	not shown (b5-verdict refuted)	b5-verdict
Step 5 live answer, USD, seconds	measured, n = 1, this run only	demo/ledger/cloud_ledger_demo.jsonl (not committed)
Step 5 recorded full-history redo of the task (USD 0.0243, 4.2 s, its answer)	measured	results/cloud_redo_full_context_v1_

B Register rows used

Current values from docs/NUMBERS_REGISTER.md, verbatim. Not used and not listed: cloud-redo-cost-per-wrong-task (the first, broken redo run) and impact-v2-P1 and impact-ranking-v2 (built on its cost per wrong task); the demo withholds them in place.

`H-scale-family-verdict` (VERIFIED, n 1 models tested)

overall alternative supported (tested: ['mistral-small3.2:24b']); mistral-small3.2:24b: R4096 0/3 (window exceeded 3/3, gated 0/3), H32768 2/3 (window exceeded 3/3); resists rule loss; history loss with rules holding holds; verdict alternative (resists rule loss)

`H-scale-verdict` (VERIFIED, n 2 models tested)

overall mixed (tested: ['qwen3:32b', 'llama3.3:70b']); qwen3:32b: R4096 2/3 (window exceeded 3/3, gated 2/3), H32768 1/3 (window exceeded 3/3); rule loss holds; does not hold; verdict mixed / qwen3-30b-a3b-2507: refused (not tested) (['qwen3-30b-a3b-2507: negative control failed (canary misses 2 at num_ctx 131072)', 'qwen3-30b-a3b-2507: task tool called correctly on 25/30 baseline turns (< 90%, task-tool gate)']) / llama3.3:70b: R4096 0/3 (window exceeded 3/3, gated 3/3), H32768 3/3 (window exceeded 3/3); resists rule loss; history loss with rules holding holds; verdict alternative (resists rule loss)

R2-mitigation-v1-verdict (VERIFIED, n 12 sessions, 480 turns)

pre-registered verdict at num_ctx_4096: refuted; num_ctx_8192 llama3.1:8b recovered; num_ctx_8192 qwen3:14b recovered; num_ctx_4096 llama3.1:8b recovered; num_ctx_4096 qwen3:14b not_recovered (rules below 90%: rule2_log_event_called)

R2-qwen3-14b-4096-reconciliation (VERIFIED, n 440 turns)

v1: rules in use 5, Ollama not recorded, harness blob not recorded, call2 off, max tokens 384, client trim None, num_keep None; per seed 20260901 rule1 40/40, rule2 40/40, rule3 40/40, rule4 20/20, rule5 39/40 / 20260902 rule1 40/40, rule2 40/40, rule3 38/40, rule4 20/20, rule5 40/40 / 20260903 rule1 40/40, rule2 40/40, rule3 37/40, rule4 20/20, rule5 39/40; gated first failure 20260901 5 rule5_session_code; 20260902 20 rule3_no_zebra; 20260903 5 rule5_session_code // v1b: rules in use 5, Ollama 0.34.4, harness blob not recorded, call2 off, max tokens 384, client trim None, num_keep None; per seed 20260904 rule1 40/40, rule2 40/40, rule3 39/40, rule4 20/20, rule5 40/40 / 20260905 rule1 40/40, rule2 40/40, rule3 40/40, rule4 20/20, rule5 40/40; gated first failure 20260904 30 rule3_no_zebra; 20260905 none // mitigation: rules in use 5, Ollama 0.34.4, harness blob not recorded, call2 off, max tokens 384, client trim {'margin': 0.05}, num_keep None; per seed 20260901 rule1 40/40, rule2 24/40, rule3 40/40, rule4 20/20, rule5 39/40 / 20260902 rule1 40/40, rule2 25/40, rule3 40/40, rule4 20/20, rule5 40/40 / 20260903 rule1 40/40, rule2 25/40, rule3 39/40, rule4 20/20, rule5 39/40; gated first failure n/a (not a real-mode run) // num_keep: rules in use 5, Ollama 0.34.4, harness blob not recorded, call2 off, max tokens 384, client trim None, num_keep system; per seed 20260901 rule1 40/40, rule2 40/40, rule3 40/40, rule4 20/20, rule5 39/40 / 20260902 rule1 40/40, rule2 40/40, rule3 38/40, rule4 20/20, rule5 40/40 / 20260903 rule1 40/40, rule2 40/40, rule3 37/40, rule4 20/20, rule5 39/40; gated first failure 20260901 5 rule5_session_code; 20260902 20 rule3_no_zebra; 20260903 5 rule5_session_code // num_keep vs v1 on the same seeds: identical final text 120/120, identical first-call prompt tokens 120/120 // first trimmed rule2 failure seed 20260901 turn 3: call 1 tools trimmed ['lookup_fact'], untrimmed v1 ['log_event', 'lookup_fact']

R2-real-v1b-gated-kill (VERIFIED, n 75)

gated killed=False; ollama_default 3/15 silent, 0 after window exceeded; num_ctx_32768 1/15 silent, 0 after window exceeded; num_ctx_16384 2/15 silent, 2 after window exceeded; num_ctx_8192 8/15 silent, 8 after window exceeded; num_ctx_4096 10/15 silent, 10 after window exceeded (llama3.1:8b excludes nothing; qwen3:14b excludes tool_args, recall; qwen3:8b excludes tool_args, recall)

R2-v1b-gated-per-model (VERIFIED, n 75)

ollama_default llama3.1:8b: 3/5 silent, 0 after window exceeded, 3 baseline-type; intact 4/4/4/2/2/2/1/1;
ollama_default qwen3:14b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/5/5/5/0/0/0;
ollama_default qwen3:8b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/5/4/4/0/0/0;
num_ctx_32768 llama3.1:8b: 1/5 silent, 0 after window exceeded, 1 baseline-type; intact 5/5/5/3/3/0/0/0;
num_ctx_32768 qwen3:14b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/5/5/0/0/0/0;
num_ctx_32768 qwen3:8b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/5/4/0/0/0/0;
num_ctx_16384 llama3.1:8b: 2/5 silent, 2 after window exceeded, 0 baseline-type; intact 5/5/0/0/0/0/0/0;
num_ctx_16384 qwen3:14b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/0/0/0/0/0/0;
num_ctx_16384 qwen3:8b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/5/0/0/0/0/0/0;
num_ctx_8192 llama3.1:8b: 4/5 silent, 4 after window exceeded, 0 baseline-type; intact 5/0/0/0/0/0/0/0;
num_ctx_8192 qwen3:14b: 0/5 silent, 0 after window exceeded, 0 baseline-type; intact 5/0/0/0/0/0/0/0;
num_ctx_8192 qwen3:8b: 4/5 silent, 4 after window exceeded, 0 baseline-type; intact 3/0/0/0/0/0/0/0;
num_ctx_4096 llama3.1:8b: 5/5 silent, 5 after window exceeded, 0 baseline-type; intact 0/0/0/0/0/0/0/0;
num_ctx_4096 qwen3:14b: 4/5 silent, 4 after window exceeded, 0 baseline-type; intact 0/0/0/0/0/0/0/0;
num_ctx_4096 qwen3:8b: 1/5 silent, 1 after window exceeded, 0 baseline-type; intact 0/0/0/0/0/0/0/0

T2S-vs-X2-default-ctx (VERIFIED, n 3)

{"qwen3-4b-2507": {"t2s_default_ctx": 4096, "x2_default_ctx": 262144}, "llama3.1:8b": {"t2s_default_ctx": 4096, "x2_default_ctx": 131072}, "qwen3:8b": {"t2s_default_ctx": 4096, "x2_default_ctx": 40960}}

b5-fa-control (VERIFIED, n 6 sessions)

llama3.1:8b: (i) f16+FA 43.4 min per session (per seed 20260901: 44.7, 20260902: 42.9, 20260903: 42.6), (g)/(h) 6.8 min, (a) 43.7 min, window 131072; rule 1.0, hist 0.9583333333333334; base-env OLLAMA_FLASH_ATTENTION ['false']; valid True; reading: the B5 speedup is KV precision, not flash attention / qwen3:14b: (i) f16+FA 206.6 min per session (per seed 20260901: 215.3, 20260902: 212.7, 20260903: 191.7), (g)/(h) 75.4 min, (a) 214.8 min, window

40960; rule 1.0, hist 0.625; base-env OLLAMA_FLASH_ATTENTION ['false']; valid True; reading: the B5 speedup is KV precision, not flash attention (tolerance 10%)

`b5-kv-recall` (VERIFIED, n 1 platform files)

evo-x2 llama3.1:8b: f16 ctx 131072 recall 50.0% (2 calls, truncated 0, errors 1; KV check n/a); q4_0 ctx 131072 recall 58.3% (3 calls, truncated 0, errors 0; KV check True); q8_0 ctx 131072 recall 54.2% (3 calls, truncated 0, errors 0; KV check True) / evo-x2 qwen3:14b: f16 ctx 40960 recall 100.0% (3 calls, truncated 0, errors 0; KV check n/a); q4_0 ctx 40960 recall 100.0% (3 calls, truncated 0, errors 0; KV check True); q8_0 ctx 40960 recall 100.0% (3 calls, truncated 0, errors 0; KV check True) / evo-t2s: not run yet / blade: not run yet

`b5-policy-metrics` (VERIFIED, n 1 platform files)

evo-x2 llama3.1:8b: a rule 100.0% hist 95.8% 43.7 min $0.0000; b rule 100.0% hist 95.8% 43.7 min $0.0000 (=a); c rule 100.0% hist 95.8% 43.6 min $0.0000; d rule 100.0% hist 95.8% 43.5 min $0.0000; e rule 100.0% hist 95.8% 43.7 min $0.0000 (=a); f rule 100.0% hist 95.8% 42.6 min $0.0000; g rule 100.0% hist 95.8% 6.9 min $0.0000; h rule 100.0% hist 100.0% 6.6 min $0.0000; frontier ['h']; complete True / evo-x2 qwen3:14b: a rule 100.0% hist 62.5% 214.8 min $0.0000; b rule 100.0% hist 62.5% 214.8 min $0.0000 (=a); c rule 100.0% hist 62.5% 178.7 min $0.0000; d rule 100.0% hist 62.5% 135.1 min $0.8419; e rule 100.0% hist 62.5% 214.8 min $0.0000 (=a); f rule 100.0% hist 62.5% 207.8 min $0.0000; g rule 100.0% hist 62.5% 75.4 min $0.0000; h rule 100.0% hist 62.5% 75.4 min $0.0000; frontier ['g']; complete True / evo-t2s: not run yet / blade: not run yet

`b5-verdict` (VERIFIED, n 3 platforms)

refuted (router dominated in evo-x2/llama3.1:8b, evo-x2/qwen3:14b); evo-x2 complete (router on frontier in every complete cell: False, fixed on frontier in every cell: []); evo-t2s not run (router on frontier in every complete cell: None, fixed on frontier in every cell: None); blade not run (router on frontier in every complete cell: None, fixed on frontier in every cell: None)

`cloud-cost-per-real-task` (VERIFIED, n 636 trajectories)

USD per real agent task, median / p90, priced, not executed (no generation call): nebius-rebench-openhands (n=36 tasks, median 62 steps; Anthropic tokens = qwen x 1.523; billed input tokens median 2,622,936 p90 4,602,617; billed output median 25,535 p90 38,481, median 183 per step): claude-haiku-5-5: none $0.2731 / $0.5063, realistic $0.1354 / $0.2454, best $0.0495 / $0.0815; claude-sonnet-5-5: none $5.4619 / $9.5990, realistic $2.5307 / $4.3387, best $0.7182 / $1.0996 | SWE-Gym (n=600 tasks, median 6 steps; Anthropic tokens = qwen x 1.506; billed input tokens median 14,404 p90 636,357; billed output median 707 p90 7,784, median 136 per step): claude-haiku-5-5: none $0.0020 / $0.0674, realistic $0.0014 / $0.0353, best $0.0009 / $0.0145; claude-sonnet-5-5: none $0.0405 / $1.3481, realistic $0.0266 / $0.6620, best $0.0173 / $0.2340; cases: none = all input uncached, realistic = ASSUMPTION cache hit rate 0.7 of the re-sent prefix (rest written at the 5-minute cache-write premium), best = every re-sent prefix read from cache, only new tokens written; each step re-sends its full prompt; output tokens per step MEASURED from the traces (assistant content + tool calls per step, results/traces/agent_step_outputs.parquet); token calibration Anthropic/qwen from messages.count_tokens on the final-step prompt of 20 sampled tasks per set (results/anthropic_token_calibration_20261021.jsonl): nebius-rebench-openhands median 1.523 (range 1.430 to 1.588, p10 1.455 p90 1.570, n=20); all median 1.511 (range 1.370 to 1.609, p10 1.457 p90 1.559, n=40); SWE-Gym median 1.506 (range 1.370 to 1.609, p10 1.474 p90 1.546, n=20); prices results/cloud_prices_anthropic_20261014.json Standard

`cloud-redo-full-context` (VERIFIED, n 197 tasks run of 197)

model claude-sonnet-5-5, thinking {'type': 'between_tools'}, full client-side history, caching on; part 1: max_tokens 384; part 2: max_tokens 2048 (max_tokens 2048 per cloud call vs the local run's 384); part 3: max_tokens 2048 (max_tokens 2048 per cloud call vs the local run's 384); 197 tasks run (34 kept from part 1, 7 part-1 rows superseded and rerun); fixed 167/197 (84.8%) of the wrong tasks, 95% Wilson CI over tasks [79.1%, 89.1%] (tasks not run count as not fixed); USD per task (cache-priced) mean $0.0227, median $0.0169, total $4.4721 (ledger basis, all input uncached: $16.8614); seconds per task mean 6.54, median 5.89; per tier: ollama_default fixed 15/17 [65.7%, 96.7%], USD mean $0.0409 median $0.0320; ollama_ctx_16384 fixed 18/19 [75.4%, 99.1%], USD mean $0.0355 median $0.0329; ollama_ctx_8192 fixed 45/53 [72.9%, 92.1%], USD mean $0.0205 median $0.0181; ollama_ctx_4096 fixed 53/63 [73.2%, 91.1%], USD mean $0.0169 median $0.0139; ollama_ctx_4096_numkeep fixed 36/45 [66.2%, 89.1%], USD mean $0.0211 median $0.0143; stops: part 1: call ended at max_tokens with nothing visible (cust_phone_info seed 20260901 turn 15); run stopped; part 2: no run_end (crashed; finished rows kept); part 3: completed; ledger total at the last run_end $38.9145; still wrong by class: thinking_empty_at_cap 0, earlier_state_repair 28, tool_round_cap 0, grader_issue 0, missing_info 0, cloud_model_error 2

cost-cut-config-time (VERIFIED, n 2 models)

MEASURED; evo-x2 llama3.1:8b: default (a) 43.7 min per session, rule 100.0% hist 95.8%; fastest measured setting (h) kv_q4_0 6.6 min, rule 100.0% hist 100.0%, window 131072; saved 37.1 min per session (85%), over 3 seeds / evo-x2 qwen3:14b: default (a) 214.8 min per session, rule 100.0% hist 62.5%; fastest measured setting (g) kv_q8_0 75.4 min, rule 100.0% hist 62.5%, window 40960; saved 139.4 min per session (65%), over 3 seeds; flash-attention control included; the flash-attention control attributes the speedup to KV precision, not flash attention (b5-fa-control)

hybrid-route-at-start (VERIFIED, n 4 device x tier rows)

route at session start: local if local accuracy at the recommended setting >= all-cloud accuracy - tolerance (points; default 0 = matched quality, 5, 10 as a sensitivity), else the whole session to the cloud at the fresh all-cloud cost (all-cloud-b3-cost) and accuracy (all-cloud-b3-quality); savings USD/month at 100 / 1,000 / 10,000 tasks/day (30-day month, stated range), measured B3 v2 point, 3 shared seeds; late-escalation and naive columns from hybrid-savings-tradeoff (escalation cost interim for claude-haiku-5-5: all-cloud per-task cost (the full-history redo ran on claude-sonnet-5-5 only); measured for claude-sonnet-5-5: cloud-redo-full-context mean USD per task); evo-t2s local accuracy measured by R2 session pass rate (not B3 task accuracy) (t2s-config-comparison, qwen3:14b), its late-escalation and naive columns at the Intel CPU 4096 default, projected from evo-x2's ollama_ctx_4096 tier / evo-x2 vs claude-haiku-5-5: recommended ollama_default, default = best, local accuracy 80/90 (88.9%) [measured, B3 task accuracy] vs all-cloud 87/90 (96.7%), gap 7.8 points (routes local at tolerance >= 7.8); tolerance 0: cloud, saves 0.00 / 0.00 / 0.00, accuracy 87/90 (96.7%); tolerance 5: cloud, saves 0.00 / 0.00 / 0.00, accuracy 87/90 (96.7%); tolerance 10: local, saves 2.01 / 20.12 / 201.25, accuracy 80/90 (88.9%); late escalation, window-flag hybrid: saves 2.01 / 20.12 / 201.25, accuracy 80/90 (88.9%), escalated 0; late escalation of every wrong task (ceiling, no such detector; all-cloud outcome and cost per escalated task (interim)): saves 1.67 / 16.70 / 167.03, accuracy 90/90 (100.0%); naive local (ollama_default, default = best): saves 2.01 / 20.12 / 201.25, accuracy 80/90 (88.9%), 10 wrong answers silent; if routed local, at projected task sizes [projected: local quality on long tasks not measured] saves at 1,000/day: SWE-Gym median 41.24, nebius-rebench-openhands median 4,060.84, nebius-rebench-openhands p90 7,360.72 / evo-x2 vs claude-sonnet-5-5: recommended ollama_default, default = best, local accuracy 80/90 (88.9%) [measured, B3 task accuracy] vs all-cloud 90/90 (100.0%), gap 11.1 points (routes local at tolerance >= 11.1); tolerance 0: cloud, saves 0.00 / 0.00 / 0.00, accuracy 90/90 (100.0%); tolerance 5: cloud, saves 0.00 / 0.00 / 0.00, accuracy 90/90 (100.0%); tolerance 10: cloud, saves 0.00 / 0.00 / 0.00, accuracy 90/90 (100.0%); late escalation, window-flag hybrid: saves 30.61 / 306.10 / 3,060.99, accuracy 80/90 (88.9%), escalated 0; late escalation of every wrong task (ceiling, no such detector; measured redo outcome per task and tier, mean redo USD): saves 23.04 / 230.43 / 2,304.29, accuracy 89/90 (98.9%); naive local (ollama_default, default = best): saves 30.61 / 306.10 / 3,060.99, accuracy 80/90 (88.9%), 10 wrong answers silent; if routed local, at projected task sizes [projected: local quality on long tasks not measured] saves at 1,000/day: SWE-Gym median 798.27, nebius-rebench-openhands median 75,921.17, nebius-rebench-openhands p90 130,160.11 / evo-t2s vs claude-haiku-5-5: recommended Intel iGPU, Ollama default, loaded window 32768, local accuracy 4/5 (80.0%) [measured by R2 session pass rate (not B3 task accuracy)] vs all-cloud 87/90 (96.7%), gap 16.7 points (routes local at tolerance >= 16.7); tolerance 0: cloud, saves 0.00 / 0.00 / 0.00, accuracy 87/90 (96.7%); tolerance 5: cloud, saves 0.00 / 0.00 / 0.00, accuracy 87/90 (96.7%); tolerance 10: cloud, saves 0.00 / 0.00 / 0.00, accuracy 87/90 (96.7%); late escalation, window-flag hybrid: saves 0.04 / 0.44 / 4.41, accuracy 87/90 (96.7%), escalated 88; late escalation of every wrong task (ceiling, no such detector; all-cloud outcome and cost per escalated task (interim), projected from evo-x2's ollama_ctx_4096 tier): saves 1.06 / 10.65 / 106.47, accuracy 89/90 (98.9%); naive local (Intel CPU default, loaded window 4096, projected from evo-x2's ollama_ctx_4096 tier): saves 2.01 / 20.12 / 201.25, accuracy 53/90 (58.9%), 37 wrong answers silent; if routed local, at projected task sizes [projected: local quality on long tasks not measured] saves at 1,000/day: SWE-Gym median 41.24, nebius-rebench-openhands median 4,060.84, nebius-rebench-openhands p90 7,360.72 / evo-t2s vs claude-sonnet-5-5: recommended Intel iGPU, Ollama default, loaded window 32768, local accuracy 4/5 (80.0%) [measured by R2 session pass rate (not B3 task accuracy)] vs all-cloud 90/90 (100.0%), gap 20.0 points (routes local at tolerance >= 20.0); tolerance 0: cloud, saves 0.00 / 0.00 / 0.00, accuracy 90/90 (100.0%); tolerance 5: cloud, saves 0.00 / 0.00 / 0.00, accuracy 90/90 (100.0%); tolerance 10: cloud, saves 0.00 / 0.00 / 0.00, accuracy 90/90 (100.0%); late escalation, window-flag hybrid: saves -35.98 / -359.80 / -3,597.97, accuracy 90/90 (100.0%), escalated 88; late escalation of every wrong task (ceiling, no such detector; measured redo outcome per task and tier, mean redo USD, projected from evo-x2's ollama_ctx_4096 tier): saves 2.61 / 26.12 / 261.20, accuracy 83/90 (92.2%); naive local (Intel CPU default, loaded window 4096, projected from evo-x2's ollama_ctx_4096 tier): saves 30.61 / 306.10 / 3,060.99, accuracy 53/90 (58.9%), 37 wrong answers silent; if routed local, at projected task sizes [projected: local quality on long tasks not measured] saves at 1,000/day: SWE-Gym median 798.27, nebius-rebench-openhands median 75,921.17, nebius-rebench-openhands p90 130,160.11

hybrid-savings-tradeoff (VERIFIED, n 16 device x tier x task-size rows)

measured trade-off, savings USD/month at 100 / 1,000 / 10,000 tasks/day (30-day month, stated range) and accuracy per option; qwen3:14b B3 v2 tasks vs all-cloud on the 3 shared seeds; escalation cost interim for claude-haiku-5-5: all-cloud per-task cost (the full-history redo ran on claude-sonnet-5-5 only); measured for claude-sonnet-5-5: cloud-redo-full-context mean USD per task; projected task-size points from cloud-cost-per-real-task (realistic caching, ASSUMPTION hit rate 0.7) apply each option's measured savings share; evo-t2s projected from evo-x2's

ollama_ctx_4096 tier; the ceiling is a separate field, not an option / evo-x2 (ollama_default, default = best) [measured] vs claude-haiku-5-5, SWE-Gym median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 4.12 / 41.24 / 412.41, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 4.12 / 41.24 / 412.41, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 3.42 / 34.23 / 342.29, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-haiku-5-5, B3 v2 tasks (30 tasks x 3 seeds), median [measured; accuracy measured]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 2.01 / 20.12 / 201.25, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 2.01 / 20.12 / 201.25, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 1.67 / 16.70 / 167.03, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-haiku-5-5, nebius-rebench-openhands median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 406.08 / 4,060.84 / 40,608.41, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 406.08 / 4,060.84 / 40,608.41, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 337.04 / 3,370.42 / 33,704.17, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-haiku-5-5, nebius-rebench-openhands p90 [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 736.07 / 7,360.72 / 73,607.24, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 736.07 / 7,360.72 / 73,607.24, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 610.93 / 6,109.25 / 61,092.54, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-sonnet-5-5, SWE-Gym median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 79.83 / 798.27 / 7,982.66, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 79.83 / 798.27 / 7,982.66, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 60.09 / 600.93 / 6,009.28, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-sonnet-5-5, B3 v2 tasks (30 tasks x 3 seeds), median [measured; accuracy measured]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 30.61 / 306.10 / 3,060.99, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 30.61 / 306.10 / 3,060.99, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 23.04 / 230.43 / 2,304.29, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-sonnet-5-5, nebius-rebench-openhands median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 7,592.12 / 75,921.17 / 759,211.69, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 7,592.12 / 75,921.17 / 759,211.69, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 5,715.29 / 57,152.88 / 571,528.84, accuracy 90/90 (100.0%) / evo-x2 (ollama_default, default = best) [measured] vs claude-sonnet-5-5, nebius-rebench-openhands p90 [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 13,016.01 / 130,160.11 / 1,301,601.08, accuracy 80/90 (88.9%), 10 wrong answers silent; hybrid flag saves 13,016.01 / 130,160.11 / 1,301,601.08, accuracy 80/90 (88.9%), escalated 0, 10 wrong answers silent; ceiling, no such detector: saves 9,798.35 / 97,983.55 / 979,835.48, accuracy 90/90 (100.0%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-haiku-5-5, SWE-Gym median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over, projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 4.12 / 41.24 / 412.41, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves 0.09 / 0.90 / 9.04, accuracy 87/90 (96.7%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 2.18 / 21.82 / 218.18, accuracy 89/90 (98.9%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-haiku-5-5, B3 v2 tasks (30 tasks x 3 seeds), median [measured; accuracy projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 2.01 / 20.12 / 201.25, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves 0.04 / 0.44 / 4.41, accuracy 87/90 (96.7%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 1.06 / 10.65 / 106.47, accuracy 89/90 (98.9%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-haiku-5-5, nebius-rebench-openhands median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over, projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 406.08 / 4,060.84 / 40,608.41, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves 8.90 / 89.05 / 890.49, accuracy 87/90 (96.7%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 214.84 / 2,148.37 / 21,483.73, accuracy 89/90 (98.9%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-haiku-5-5, nebius-rebench-openhands p90 [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over, projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 87/90 (96.7%); naive local saves 736.07 / 7,360.72 / 73,607.24, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves 16.14 / 161.41 / 1,614.12, accuracy 87/90 (96.7%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 389.42 / 3,894.16 / 38,941.63, accuracy 89/90 (98.9%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-sonnet-5-5, SWE-Gym median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over, projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 79.83 / 798.27 / 7,982.66, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves -93.83 / -938.30 / -9,383.02, accuracy 90/90 (100.0%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 6.81 / 68.12 / 681.18, accuracy 90/90 (100.0%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-sonnet-5-5, B3 v2 tasks (30 tasks x 3 seeds), median [measured; accuracy projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 30.61 / 306.10 / 3,060.99, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves -35.98 / -359.80 / -3,597.97, accuracy 90/90 (100.0%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 2.61 / 26.12 / 261.20, accuracy 90/90

(100.0%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-sonnet-5-5, nebius-rebench-openhands median [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over, projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 7,592.12 / 75,921.17 / 759,211.69, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves -8,923.97 / -89,239.74 / -892,397.43, accuracy 90/90 (100.0%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 647.85 / 6,478.51 / 64,785.13, accuracy 90/90 (100.0%) / evo-t2s Intel CPU (default, loaded window 4096) [projected] vs claude-sonnet-5-5, nebius-rebench-openhands p90 [projected: local quality on long tasks not measured; accuracy B3 accuracy carried over, projected from evo-x2's ollama_ctx_4096 tier]: all-cloud saves 0, accuracy 90/90 (100.0%); naive local saves 13,016.01 / 130,160.11 / 1,301,601.08, accuracy 53/90 (58.9%), 37 wrong answers silent; hybrid flag saves -15,299.36 / -152,993.62 / -1,529,936.21, accuracy 90/90 (100.0%), escalated 88, 0 wrong answers silent; ceiling, no such detector: saves 1,110.68 / 11,106.84 / 111,068.35, accuracy 90/90 (100.0%)

`impact-P3` (VERIFIED, n 3)

claude-sonnet-5-5 standard caching on, sessions shaped like llama3.1:8b's: status measured; measured USD/day 3.60, assumed USD/day 0.00, range USD/day 3.60 to 71.96; per session: re-sent history 1,798,884 tok, USD 0.1799 caching on, 3.5978 no caching; whole session to the cloud USD 0.3358 caching on, 3.7302 no caching (n=5); assumptions: team size = 10 (assumption (profile JSON)); agent sessions per developer per day = 4 (assumption (profile JSON)); cloud session share = 0.5 (assumption (profile JSON)); cloud model = claude-sonnet-5-5 standard, caching on (results/cloud_prices_anthropic_20261014.json (official Anthropic pricing page)); token basis = o200k_base counts (assumption: Anthropic tokenizer counts not measured); re-sent history is a cacheable prefix = R2 sessions are append-only (best case for caching; real histories not measured); confidence high

`impact-hour-value` (VERIFIED, n 1)

USD 64.44/h ($64.44 per hour; Quick Facts: Software Developers, Quality Assurance Analysts, and Testers, row "2025 Median Pay"; May 2025; https://www.bls.gov/ooh/computer-and-information-technology/software-developers.htm; fetched 2026-10-14)

`impact-total` (VERIFIED, n 7)

status quo ollama_default, claude-sonnet-5-5 caching on: measured-only USD/day 3.60 (USD/month 75.55); range USD/day 3.60 to 192.81 (USD/month 75.55 to 4,049.08; low = measured only, nobody waits for compute; high = compute time at the full hour value and cloud without caching) over P2, P3, P4, P5, P6, P7; not added (assumed or not measured): P1; ranking (caching on, time at its upper bound) P5 > P6 > P2 > P3 > P4 > P7 > P1; per row status P1 assumed, P2 mixed, P3 measured, P4 mixed, P5 mixed, P6 mixed, P7 measured

`late-escalation-cost-vs-fresh` (VERIFIED, n 197 redo tasks, 90 fresh tasks, 119 matched rows)

claude-sonnet-5-5, USD per task (cache-priced): late full-history escalation of a wrong local task (cloud-redo-full-context) mean $0.0227, median $0.0169 (n=197) vs a fresh all-cloud run (all-cloud-b3-cost) mean $0.0102, median $0.0090 (n=90); ratio 2.22x on the mean, 1.87x on the median; matched on (seed, task_id) where both exist (seeds 20260901, 20260902, 20260903): 119 redo rows over 53 distinct tasks, redo mean $0.0226, median $0.0169 (n=119) vs fresh of the same task mean $0.0129, median $0.0103 (n=119), ratio 1.76x mean, 1.64x median; incomplete: fixed 167/197 (84.8%), 95% Wilson CI 79.1% to 89.1%; of the 30 unfixed, 28 inherited the local model's earlier executed actions (earlier_state_repair), cloud_model_error 2; evo-t2s Intel CPU (default, loaded window 4096) [projected, from evo-x2's ollama_ctx_4096 tier] vs claude-sonnet-5-5 (hybrid-savings-tradeoff, B3 v2 tasks): the window-flag hybrid escalates 88 of 90 tasks and saves USD -35.98 / -359.80 / -3,597.97/month at 100 / 1,000 / 10,000 tasks/day, accuracy 90/90 (100.0%) vs all-cloud 90/90 (100.0%): it costs more than all-cloud

`num-keep-guardrail-test` (VERIFIED, n 120 turns)

llama3.1:8b num_ctx 4096 num_keep=system: S per seed {20260901: 601, 20260902: 594, 20260903: 597}; observed context shifts 105 with n_keep [595, 598, 602], task n_keep [594, 597, 601], token cuts 7 with keep [594, 601], –keep [], Ollama message-drop calls 224 of 239; rules in use: rule1 94.2% (n=120, first failure 31/-/5); rule3 100.0% (n=120, first failure -/-/-); rule4 100.0% (n=60, first failure -/-/-); canary misses C 6, H 24 of 24 checks; error turns 2; verdict: one-line fix

`num-keep-qwen3-14b` (VERIFIED, n 120 turns)

qwen3:14b num_ctx 4096 num_keep=system: S per seed {20260901: 752, 20260902: 747, 20260903: 750}; observed context shifts 0 with n_keep [], task n_keep [747, 750, 752], token cuts 0 with keep [], –keep [], Ollama message-drop

calls 228 of 240; rules in use: rule1 100.0% (n=120, first failure -/-/-); rule2 100.0% (n=120, first failure -/-/-); rule3 95.8% (n=120, first failure -/20/20); rule4 100.0% (n=60, first failure -/-/-); rule5 98.3% (n=120, first failure 5/-/5); canary misses C 1, H 24 of 24 checks; error turns 0; verdict: inconclusive (flag effect not observed)

`t2s-config-comparison` (VERIFIED, n 4 device x model rows)

evo-t2s configuration, measured R2 data (evo-t2s R2 real, 5 seeds x 40 turns); to be replaced by the evo-t2s B5 results (2026-10-26 to 2026-11-08) when they land; kept local, measured by R2 session pass rate (not B3 task accuracy) = share of R2 sessions with no gated silent failure (t2s-r2-real-cpu, t2s-r2-real-igpu); savings = B x k - X with B the measured all-cloud B3 per-task cost and X = (1 - k) x B x (escalation USD per escalated task / all-cloud USD per task - 1): 0 for a tier whose escalated task is costed at its all-cloud price (interim), positive for a tier costed at the measured full-history redo price (see hybrid-savings-tradeoff for which applies) (ASSUMPTION: R2 session share stands for the task share); 30-day month; naive: Intel CPU, Ollama default llama3.1:8b: loaded window 4096, gated silent 5/5 sessions, kept local (measured by R2 session pass rate (not B3 task accuracy)) 0/5 = 0.00; vs claude-haiku-5-5: saves USD 0.00 at 100/day, 0.00 at 1,000/day, 0.00 at 10,000/day/month; vs claude-sonnet-5-5: saves USD -37.49 at 100/day, -374.93 at 1,000/day, -3,749.31 at 10,000/day/month / naive: Intel CPU, Ollama default qwen3:14b: loaded window 4096, gated silent 5/5 sessions, kept local (measured by R2 session pass rate (not B3 task accuracy)) 0/5 = 0.00; vs claude-haiku-5-5: saves USD 0.00 at 100/day, 0.00 at 1,000/day, 0.00 at 10,000/day/month; vs claude-sonnet-5-5: saves USD -37.49 at 100/day, -374.93 at 1,000/day, -3,749.31 at 10,000/day/month / recommended: Intel iGPU, Ollama default llama3.1:8b: loaded window 32768, gated silent 0/5 sessions, kept local (measured by R2 session pass rate (not B3 task accuracy)) 5/5 = 1.00; vs claude-haiku-5-5: saves USD 2.01 at 100/day, 20.12 at 1,000/day, 201.25 at 10,000/day/month; vs claude-sonnet-5-5: saves USD 30.61 at 100/day, 306.10 at 1,000/day, 3,060.99 at 10,000/day/month / recommended: Intel iGPU, Ollama default qwen3:14b: loaded window 32768, gated silent 1/5 sessions, kept local (measured by R2 session pass rate (not B3 task accuracy)) 4/5 = 0.80; vs claude-haiku-5-5: saves USD 1.61 at 100/day, 16.10 at 1,000/day, 161.00 at 10,000/day/month; vs claude-sonnet-5-5: saves USD 16.99 at 100/day, 169.89 at 1,000/day, 1,698.93 at 10,000/day/month

`t2s-r2-mitigation-cpu` (VERIFIED, n 6 sessions, 240 turns)

overall refuted; num_ctx_4096 llama3.1:8b recovered; num_ctx_4096 qwen3:14b not_recovered (rules below 90%: rule2_log_event_called)

`t2s-r2-real-cpu` (VERIFIED, n 10 sessions)

llama3.1:8b: loaded window [4096], first over window turn [3]; gated 5/5 silent, 5 after window exceeded, 0 baseline-type (gate excludes recall); intact 0/0/0/0/0/0/0/0 / qwen3:14b: loaded window [4096], first over window turn [3]; gated 5/5 silent, 5 after window exceeded, 0 baseline-type (gate excludes recall); intact 0/0/0/0/0/0/0/0; pre-registered kill criterion killed=False, ollama_default 10/10 silent

`t2s-r2-real-igpu` (VERIFIED, n 10 sessions)

llama3.1:8b: loaded window [32768], first over window turn [23]; gated 0/5 silent, 0 after window exceeded, 0 baseline-type (gate excludes nothing); intact 5/5/5/4/4/0/0/0 / qwen3:14b: loaded window [32768], first over window turn [20, 21]; gated 1/5 silent, 0 after window exceeded, 1 baseline-type (gate excludes recall); intact 4/4/4/4/0/0/0/0; pre-registered kill criterion killed=False, ollama_default 5/10 silent

`x2-evalset-tasks-real-v2-ci-units` (VERIFIED, n 25 sessions, 750 task turns)

qwen3:14b wrong-task excess over default; ollama_ctx_16384: excess +1.3 points; session-level 95% CI -1.3 to +4.0 (5 vs 5 sessions); turn-level 95% CI -6.0 to +8.7 (150 vs 150 task turns); ollama_ctx_8192: excess +24.0 points; session-level 95% CI +18.7 to +29.3 (5 vs 5 sessions); turn-level 95% CI +14.7 to +33.3 (150 vs 150 task turns); ollama_ctx_4096: excess +30.7 points; session-level 95% CI +26.0 to +34.7 (5 vs 5 sessions); turn-level 95% CI +21.3 to +40.0 (150 vs 150 task turns); ollama_ctx_4096_numkeep: excess +18.7 points; session-level 95% CI +16.0 to +21.3 (5 vs 5 sessions); turn-level 95% CI +10.0 to +27.3 (150 vs 150 task turns) (10000 resamples each, seed 20261014; the paper uses the session-level CI)

`x2-evalset-tasks-real-v2-mistral-small32-wrong-rate` (PENDING, n n/a)

not computed: data file(s) not synced yet: results/x2_evalset_tasks_real_v2_mistral_small32.jsonl

`x2-evalset-tasks-real-v2-qwen3-8b-wrong-rate` (VERIFIED, n 0 task turns)

run finished False; qwen3:8b: refused (qwen3:8b: task tool called correctly on 61/81 baseline turns (< 90%, task-tool gate); qwen3:8b: task end-state gate failed (65/90 < 90%); qwen3:8b: strict preflight, gates not passed: ['baseline', 'task_tool', 'task_end_state']) (bootstrap 10000 resamples, unit session, stratified by tier, seed 20261014)

`x2-evalset-tasks-real-v2-wrong-rate` (VERIFIED, n 750 task turns)

run finished True; llama3.1:8b: refused (llama3.1:8b: negative control failed (canary misses 26 at num_ctx 131072); llama3.1:8b: not in this plan's model list ['qwen3:14b']; llama3.1:8b: task end-state gate failed (72/90 < 90%)) / qwen3:14b: valid True; ollama_default: wrong 17/150 (11.3%) (silent 17, surfaced 0), loaded context [40960]; ollama_ctx_16384: wrong 19/150 (12.7%) (silent 19, surfaced 0), loaded context [16384]; ollama_ctx_8192: wrong 53/150 (35.3%) (silent 53, surfaced 0), loaded context [8192]; ollama_ctx_4096: wrong 63/150 (42.0%) (silent 63, surfaced 0), loaded context [4096]; ollama_ctx_4096_numkeep: wrong 45/150 (30.0%) (silent 45, surfaced 0), loaded context [4096], num_keep per seed {20260901: 3369, 20260902: 3369, 20260903: 3364, 20260904: 3369, 20260905: 3367}; ollama_ctx_16384 excess over default +1.3 points (95% CI -1.3 to +4.0, 5 vs 5 sessions); reading: no excess shown; ollama_ctx_8192 excess over default +24.0 points (95% CI +18.7 to +29.3, 5 vs 5 sessions); reading: ollama_ctx_8192 raises the wrong-task rate; ollama_ctx_4096 excess over default +30.7 points (95% CI +26.0 to +34.7, 5 vs 5 sessions); reading: ollama_ctx_4096 raises the wrong-task rate; ollama_ctx_4096_numkeep excess over default +18.7 points (95% CI +16.0 to +21.3, 5 vs 5 sessions); reading: ollama_ctx_4096_numkeep raises the wrong-task rate (bootstrap 10000 resamples, unit session, stratified by tier, seed 20261014)

`x2-scale-real-default-tier` (VERIFIED, n 3 sessions)

llama3.3:70b: no default-tier session / qwen3:32b: 3 default-tier sessions completed (seed 20260901: 40 turns, 6.18 h, loaded [40960], first over window 24; seed 20260902: 40 turns, 6.01 h, loaded [40960], first over window 25; seed 20260903: 40 turns, 4.66 h, loaded [40960], first over window 26)

`x2-v3-usability` (VERIFIED, n 121)

per-call timeout 900 s (harness DEFAULT_CALL_TIMEOUT_S); timeout_latency/done (share) [context tiers]: llama3.1:8b/llama_server 40/340 (11.8%) [longdoc_120000 x10, longdoc_64000 x10, longdoc_80000 x10, longdoc_96000 x10]; llama3.1:8b/ollama_default 10/340 (2.9%) [longdoc_120000 x10]; qwen3-14b/llama_server 0/340 (0.0%); qwen3-14b/ollama_default 0/340 (0.0%); qwen3-30b-a3b/llama_server 10/340 (2.9%) [longdoc_120000 x10]; qwen3-30b-a3b/ollama_default 0/340 (0.0%); qwen3-32b/llama_server 11/100 (11.0%) [longdoc_32000 x10, trace_mix_24000 x1]; qwen3-32b/ollama_default 0/100 (0.0%); qwen3-4b-2507/llama_server 40/340 (11.8%) [longdoc_120000 x10, longdoc_64000 x10, longdoc_80000 x10, longdoc_96000 x10]; qwen3-4b-2507/ollama_default 10/340 (2.9%) [longdoc_120000 x10]; qwen3-8b/llama_server 0/340 (0.0%); qwen3-8b/ollama_default 0/340 (0.0%)